

Section 13.1 Functions of Two or More Variables

Topics: (1) Definition of Functions

(2) Finding Domains

(3) Graphing Functions of Two Variables

(4) Level Curves

Recall: Functions of Single Variable $y = f(x)$, $x = f(y)$

$y = x^2$: Input = y . Then it's not a function of y , because if $y = 4$, then $x = \pm 2$.

But $y = x^2$: Input = x . Then it's a function of x . So, $y = f(x) = x^2$.

Note: In a function, output cannot have an even power.

Recall: Definition: For any real number x , $\sqrt{x^2} = |x|$. For example, $\sqrt{16} = 4$

Writing a function from a non-function:

$$y = x^2 \Rightarrow x = \pm\sqrt{y}$$

Note that $\sqrt{9} = 3$ and $-\sqrt{9} = -3$

Positive Square Root: $x = \sqrt{y}$. Here $x \geq 0$. $D = [0, \infty)$ and $R = [0, \infty)$

Negative Square Root: $x = -\sqrt{y}$. Here $x \leq 0$. $D = [0, \infty)$ and $R = (-\infty, 0]$

Observation:

Complex Number:

$\sqrt{-1} = i$, that is, $i^2 = -1 \rightarrow$ imaginary identity, $z = a + bi \rightarrow$ imaginary/complex number.

Note that $\sqrt{-1}$, $\sqrt[4]{-1}$, $\sqrt[6]{-1}$, $\sqrt[even]{-1} \dots$ are also imaginary number.

Radical: The n^{th} root of x is denoted by $\sqrt[n]{x}$ and is called the radical.

- x is any real number
- n is a positive integer, and n is called the index of the radical.
- If n is even and if x is a negative number, then $\sqrt[n]{x}$ is an imaginary number.
- $\sqrt[n]{\text{negative}}$ is an imaginary number.

A **function** is a relation between a set of inputs and a set of permissible outputs with the property that **each input is related to exactly one output**.

$[y = f(x) = x^2, \text{ If } x = 1, \text{ then } y = 1 \text{ and if } x = -1, \text{ then } y = 1]$, this is a function, but not 1-to-1.

Definition: Let X and Y be two nonempty sets. A **function** from X into Y is a relation that associates with each element of X exactly one element of Y

Note: For a function, no input has more than one output. The **domain** of a function is the set of all inputs; the **range** is the set of all outputs.

Undefined: $\frac{\text{non-zero}}{0}$ and complex: $\sqrt[n]{-1}$. $[\frac{0}{0} = \text{indeterminant}, \frac{0}{3} = 0]$

Odd roots are not a problem, because $\sqrt[n]{-1} = (-1)^{\text{odd}} = -\sqrt[n]{1} = -1$.

For example, $(-1)^7 = -1$. Hence, $\sqrt[7]{-1} = -1$ and $\sqrt[3]{-8} = -\sqrt[3]{8} = -2$.

Note: If $x^{\text{even}} < 0$, then x is a complex number.

Domain: Collection of all inputs so that the outputs are **defined and real**.

Domain of $y = \sqrt[n]{x+1} = \frac{\sqrt[n]{x+1}}{1}$. For the domain, $x+1 \geq 0$. $D = [-1, \infty)$

Domain of $y = \sqrt[n]{x+1} = \frac{\sqrt[n]{x+1}}{1}$. $D = (-\infty, \infty)$

NOTATION AND TERMINOLOGY

There are many familiar formulas in which a given variable depends on two or more other variables. For example, the area A of a triangle depends on the base length b and height h by the formula

$$A = \frac{1}{2}bh; \rightarrow A = f(b, h), \text{ input} = (b, h)$$

the volume V of a rectangular box depends on the length l , the width w , and the height h by the formula

$$V = lwh; \quad V = f(l, w, h), \quad \text{input} = (l, w, h)$$

and the arithmetic average $\bar{x}$ of n real numbers, $x_1, x_2, \dots, x_n$, depends on those numbers by the formula

$$\bar{x} = \frac{1}{n} (x_1 + x_2 + \dots + x_n) \rightarrow \bar{x} = f(x_1, x_2, \dots, x_n), \text{ input} = (x_1, x_2, \dots, x_n)$$

Thus, we say that

A is a function of the two variables b and h ;

V is a function of the three variables l, w , and h ;

$\bar{x}$ is a function of n variables $x_1, x_2, \dots, x_n$.

The terminology and notation for functions of two or more variables is similar to that for functions of one variable. For example, the expression

$$\mathbf{z} = \mathbf{f}(\mathbf{x}, \mathbf{y}) \quad [\text{For } (x, y, z), \text{ here } z \text{ is the output} \quad]$$

means that **z is a function of x and y** in the sense that a unique value of the dependent variable z is determined by specifying values for the independent variables x and y .

Definition: A **function f of two variables**, x and y , is a **rule** that assigns a unique real number $z = f(x, y)$ to each point (x, y) in some set D in the xy —plane.

Graph $f = \{(x, y, z): z = f(x, y) \text{ for } (x, y) \in D\}$

Example: $f(x, y) = e^x \cos y$, $f(x, y) = \cos(x^2 y)$, $z = f(x, y) = 1$

Note: $y = f(x) = 1$ is a horizontal line. That means, $z = f(x, y) = 1$ is a horizontal plane.

Note: Given that $z = f(x, y) = e^x \cos y + 1$.

Projection on the xy -plane $z = 0$: $e^x \cos y + 1 = 0$, that is, $y = \cos^{-1}(-e^{-x})$

Similarly,

$$w = f(x, y, z)$$

expresses w as a function of x , y , and z .

Definition: A **function f of three variables**, x , y , and z , is a **rule** that assigns a unique real number $w = f(x, y, z)$ to each point (x, y, z) in some set D in three dimensional space.

Example: $f(x, y, z) = x^2 \cos y \sin z$, : $f(x, y, z) = x^2 \cos(xyz^3) \sin(yz)$,

$$w = f(x, y, z) = 1$$

Domain

Unless specifically stated otherwise, the domain of a function of several variables is the set of all values of the variables for which the given expression is defined (It gives real values).

Recall: Domain of a Function of a single variable $y = f(x)$

Domain of a function f is the largest set of real numbers for which the value $f(x)$ is a real number.

Finding the Domain of a Function Defined by an Equation:

1. Start with the domain as the set of all real numbers.
2. If the equation has a denominator, exclude any numbers that give a zero denominator.

3. If the equation has a radical of even index, exclude any numbers that cause the expression inside the radical to be negative. $[\sqrt{-1}, \sqrt[4]{-1}, \sqrt[6]{-1}, \dots \text{Not real}]$

Good: $\sqrt[3]{-1} = -1$, $\sqrt[3]{-8} = -2$, $\sqrt[odd]{-a} = -\sqrt[odd]{a}$ where $a > 0$, Always real. Also, $\sqrt[n]{0} = 0^{\frac{1}{n}} = 0$.

Recall: $\sqrt[n]{X}$ is a radical, called n -th root. The number n is a positive integer, called the index.

Note: $\frac{1}{0}$ is undefined. $\sqrt{-1} = i$, $\sqrt[even]{-1}$ is NOT real.

Note that $x^2 \geq 0$ for any real number x . If $\sqrt[4]{-1} = a \Rightarrow a^4 = -1$, which is not true for any real number a .

Example: Find the domain of the following functions:

1) a) $f(x) = x^2 + 4$. **Answer:** Domain = all real numbers

$$b) (1) f(x) = \frac{x^2+1}{x^2-5x+6} = \frac{x^2+1}{(x-2)(x-3)}$$

Answer: Domain = $\{x \in \mathbb{R} : x \neq 2, 3\} = (-\infty, 2) \cup (2, 3) \cup (3, \infty)$

[Note that if $x \in (2, 3)$ means $2 < x < 3$; if $x \in [2, 3]$ means $2 \leq x \leq 3$; and if $x \in [2, 3)$ means $2 \leq x < 3$]



(2) $f(x) = \frac{x^2+1}{x^2+6}$; $D = \text{All real numbers}$.

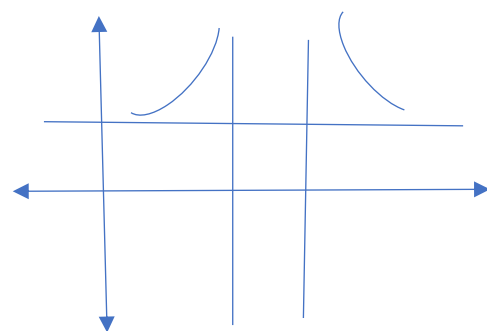
$$a) f(x) = \sqrt{2x+4} = \frac{\sqrt{2x+4}}{1}$$

Answer: For the domain, set $2x + 4 \geq 0 \Rightarrow 2x \geq -4$.

Domain = $\{x : 2x + 4 \geq 0\} = \{x : x \geq -2\} = [-2, \infty)$

(2) b) (i) $f(x) = \sqrt[5]{2x+4}$; $D = \text{All real numbers}$.

Note that $\frac{0}{1} = 0$.



b) (ii) $f(x) = \frac{5x-3}{\sqrt{3x-7}}$

Answer: Domain = $\{x : 3x - 7 > 0\} = \left\{x : x > \frac{7}{3}\right\} = \left(\frac{7}{3}, \infty\right)$.

Note: $\sqrt[n]{0} = 0$.

Note that $3x - 7 \neq 0$. If $3x - 7 = 0$, then $\sqrt{3x - 7} = 0$. Hence, $\frac{5x-3}{\sqrt{3x-7}}$ is undefined.

So, for the domain, we want $3x - 7 > 0 \Rightarrow x > \frac{7}{3}$.

$f(x) = \frac{5x-3}{\sqrt{3x-7}}$; Domain = $\left(\frac{7}{3}, \infty\right)$.

(ii) $f(x) = \frac{5x-3}{\sqrt[11]{3x-7}}$, $D = \left\{x \in \mathbb{R} : x \neq \frac{7}{3}\right\} = \left(-\infty, \frac{7}{3}\right) \cup \left(\frac{7}{3}, \infty\right)$

Sum, Difference, Product and Quotient of Two Functions:

If f and g are two functions, then

1. The **sum** $f + g$ is the function defined by $(f + g)(x) = f(x) + g(x)$

The domain of $f + g$ = domain of $f \cap$ domain of g .

2. The **difference** $f - g$ is the function defined by $(f - g)(x) = f(x) - g(x)$

The domain of $f - g$ = domain of $f \cap$ domain of g .

3. The **product** $f \cdot g$ is the function defined by $(f \cdot g)(x) = f(x) \cdot g(x)$

The domain of $f \cdot g$ = domain of $f \cap$ domain of g .

For Examples, (1) $f(x) = \sin x$, $g(x) = \ln x$.

Domain of $f = (-\infty, \infty)$ and Domain of $g = (0, \infty)$

Now, let $h(x) = (f \cdot g)(x) = f(x) \cdot g(x) = \sin x \cdot \ln x$, that is, $h(x) = \sin x \cdot \ln x$

Domain of $h = \text{Domain of } f \cap \text{Domain of } g = (0, \infty)$.

(2) **Example:** Given, $f(x) = \frac{x-2}{x-1}$, $g(x) = \frac{x}{x-2}$

$h(x) = (f \cdot g)(x) = f(x) \cdot g(x) = \frac{x-2}{x-1} \cdot \frac{x}{x-2}$, that is, $h(x) = \frac{x}{x-1}$; $x \neq 2$

4. The **quotient** $\frac{f}{g}$ is the function defined by $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$, $g(x) \neq 0$.

The domain of $\frac{f}{g} = \{x \mid g(x) \neq 0\} \cap \text{domain of } f \cap \text{domain of } g$.

Example: Given, $f(x) = \frac{1}{x-1}$, $g(x) = \frac{x}{x-2}$

Note that $g(x) = \frac{x}{x-2} = 0$ iff $x = 0$.

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{\frac{1}{x-1}}{\frac{x}{x-2}}, x \neq 1, 2 = \frac{x-2}{x(x-1)}; x \neq 1, 2$$

The domain of $\frac{f}{g} = \{x \mid g(x) \neq 0\} \cap \text{domain of } f \cap \text{domain of } g$

$$= \{x \mid x \neq 0, 1, 2\} = (-\infty, 0) \cup (0, 1) \cup (1, 2) \cup (2, \infty)$$

.....

Note that, the domain of $y = \frac{x-2}{x(x-1)}$ is

$$= (-\infty, 0) \cup (0, 1) \cup (1, \infty)$$

➔ Note: We need to know Trigonometric, Exponential and Logarithmic functions.

Exponential Function: $y = b^x$, $b > 0$, $b \neq 1$; Inverse: $y = \log_b x$

(1) If $0 < b < 1$, then $y = b^x$ is decreasing. (2) If $b > 1$, then $y = b^x$ is increasing.

Examples: (1) Consider, $y = 2^x$, $y = \pi^x$, $y = e^x$, $D = (-\infty, \infty)$, $R = (0, \infty)$

(2) (a) Consider, $y = \pi^{\sqrt{x}}$, $D = [0, \infty)$, $R = [1, \infty)$

(b) Consider, $y = \left(\frac{1}{2}\right)^{\sqrt{x}}$, $D = [0, \infty)$, $R = (0, 1]$

(3) Consider, $y = \pi^{-\sqrt{x}} = \frac{1}{\pi^{\sqrt{x}}} = \left(\frac{1}{\pi}\right)^{\sqrt{x}}$, $D = [0, \infty)$, $R = (0, 1]$

(4) $y = \tan x = \frac{\sin x}{\cos x}$; Domain = $\{x : x \neq (2n + 1)\frac{\pi}{2} \text{ for any integer } n\}$

Note: $y = \frac{x-1}{x^2-1} = \frac{1}{x+1}$, because $\frac{x-1}{x-1} = 1, x-1 \neq 0$.

$$y = \frac{x-1}{x^2-1} = \frac{1}{x+1}, x \neq 1 = \frac{1}{x+1} \cdot \frac{x-1}{x-1}.$$

Exercises from MAT-250

Exercise: 1 Consider the function $f(x, y) = \sqrt{x+1} - \log_2(4 - x^2 - y^2)$.

(a) Evaluate $f(0, 0)$, $f(0, \sqrt{2})$, $f(3, -1)$ and $f(1, -1)$

(b) Find the Domain of the function f , and **sketch it**.

$$\text{Solution: } f(0, 0) = \sqrt{1} - \log_2 4 = 1 - \log_2 2^2 = 1 - 2 = -1,$$

$$f(0, \sqrt{2}) = 1 - 1 = 0$$

$$\text{and } f(1, -1) = \sqrt{2} - \log_2 2 = \sqrt{2} - 1$$

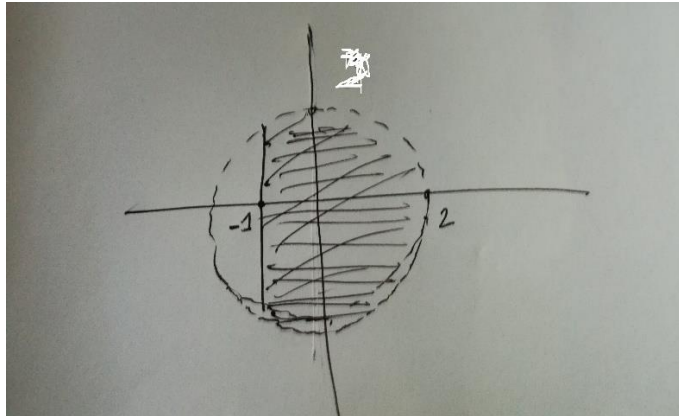
Note that $f(3, -1)$ is not defined, because -6 is not in the domain of the log function.

For the domain of $f(x, y) = \sqrt{x+1} - \log_2(4 - x^2 - y^2)$:

We need to have $x+1 \geq 0$ and $4 - x^2 - y^2 > 0$

That is, $x \geq -1$ and $x^2 + y^2 < 2^2$ [note that $x^2 + y^2 = 2^2$ is a circle]

Domain: $\{(x, y) : x \geq -1 \quad \text{and} \quad x^2 + y^2 < 2^2\}$



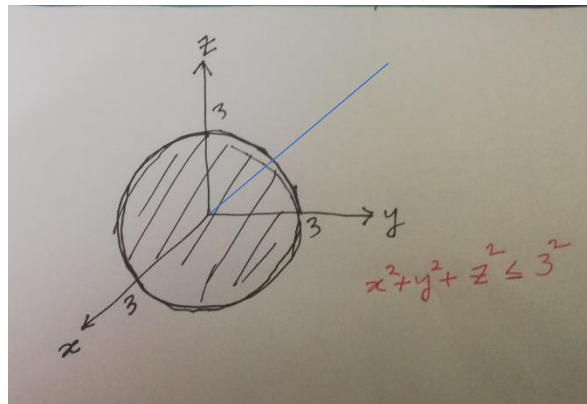
Exercise: 2A Consider the function $f(x, y, z) = \sqrt{9 - x^2 - y^2 - z^2}$

(a) Evaluate $f(0, 0, 0)$, $f(0, \sqrt{2}, 2)$ and $f(1, 3, 0)$

→ Since $f(1, 3, 0)$ is not real, hence $(1, 3, 0)$ is not in the domain.

(b) Find the Domain of the function f and sketch it.

For the domain, set $9 - x^2 - y^2 - z^2 \geq 0 \Rightarrow x^2 + y^2 + z^2 \leq 3^2$



Exercise: 2B Consider the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2 - 4}$

Exercise: 2C Consider the function $f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2 - 4}}$

Exercise: 2D Consider the function $f(x, y, z) = \frac{2x+y^2-1}{\sqrt{4-(x^2+y^2+z^2)}}$

Definition: The domain of the function $y = (f \circ g)(x) = f(g(x))$ is

$$D = \{x : g(x) \in \text{Domain of } f\}$$

Exercise: 3 Consider the function $f(x, y) = \sin\left(\frac{x+y}{\sqrt{x^2-y}}\right)$

[note that if $z = \frac{x+y}{\sqrt{x^2-y}}$ is real, then it will be in the domain of the *sine* function]

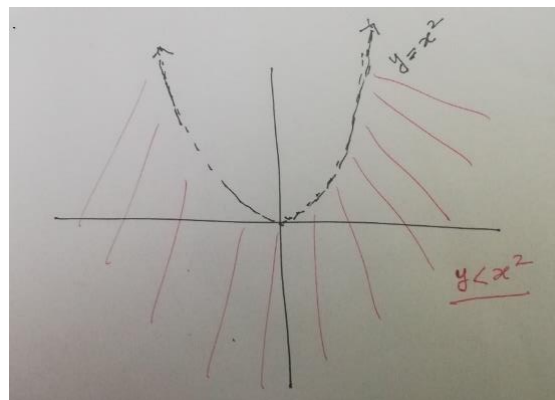
(a) Evaluate $f(0, 0)$, $f(\sqrt{2}, 0)$ and $f(1, -1)$

(b) Find the Domain of the function f .

$$\text{Domain} = \{(x, y) : \frac{x+y}{\sqrt{x^2-y}} \text{ is a real number}\}$$

For the domain of the function $x^2 - y > 0$, i.e., $y < x^2$. [$y = f(x)$]

$$\text{Domain} = \{(x, y) : y < x^2\}$$



Exercise: 4 (a) Consider the function $f(x, y) = \sin\left(\frac{x+y}{\sqrt{x^2-y}}\right)$

$$D = \{(x, y) \in \mathbb{R}^2 : y \neq x^2\}.$$

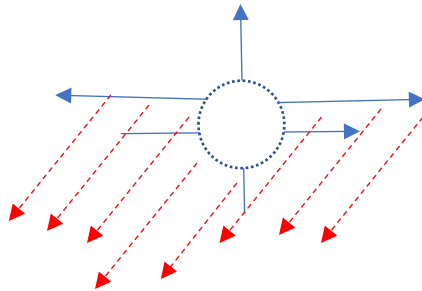
(b) Consider the function $f(x, y) = \sin(\sqrt[9]{x^2 - y})$; Domain = The whole xy -plane.

Exercise: 5a Consider the function $f(x, y) = \sqrt[3]{x+1} - e^{\sqrt{2-y}} + \ln \sqrt{x^2 + y^2 - 9}$.

Solution: $g(x, y) = \sqrt[3]{x+1} = \frac{\sqrt[3]{x+1}}{1}$; $D = \text{All real numbers.}$

$$h(x, y) = e^{\sqrt{2-y}} ; D = \{(x, y): 2 - y \geq 0\} = \{(x, y): y \leq 2\}$$

$$k(x, y) = \ln \sqrt{x^2 + y^2 - 9} ; D: x^2 + y^2 > 9$$



Exercise: 5b Consider the function $f(x, y) = \sqrt[4]{2-x} - e^{\sqrt[3]{2-y}} + \ln \sqrt{x^2 + y^2 - 9}$.

Find the domain and sketch it.

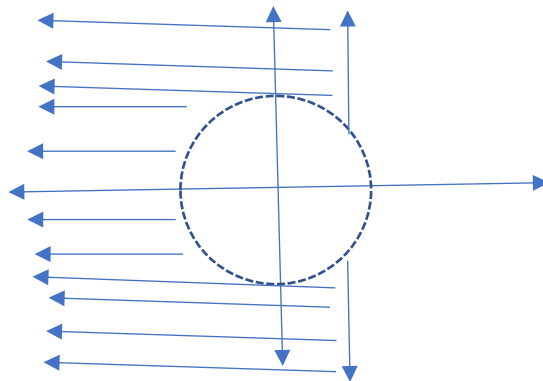
Solution: For the function $z = \sqrt[3]{2-y}$, z can be anything

For the function $w = e^{\sqrt[3]{2-y}}$, w can be anything. Hence the exponential function does not change the domain of the function f .

For the domain we get, $2 - x \geq 0$ and $x^2 + y^2 - 9 > 0$, that is,

$$x \leq 2 \text{ and } x^2 + y^2 > 3^2$$

Domain = $\{(x, y) | x \leq 2 \text{ and } x^2 + y^2 > 3^2\}$.



The domain is the region that is outside the circle $x^2 + y^2 = 3^2$ and to the left of the vertical line $x = 2$.

Exercise: 6 Consider the function $f(x, y) = \sqrt[3]{x+1} - \sqrt[4]{1-y} + \ln \sqrt{x^2 + y^2 - 4}$.

Solution: $z = \sqrt[3]{x+1}$ $D = \text{All real numbers}$,

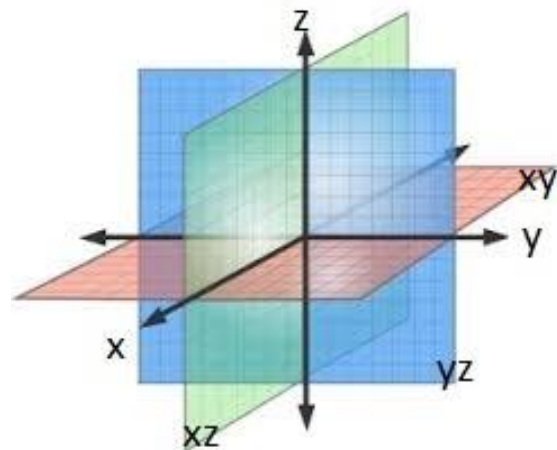
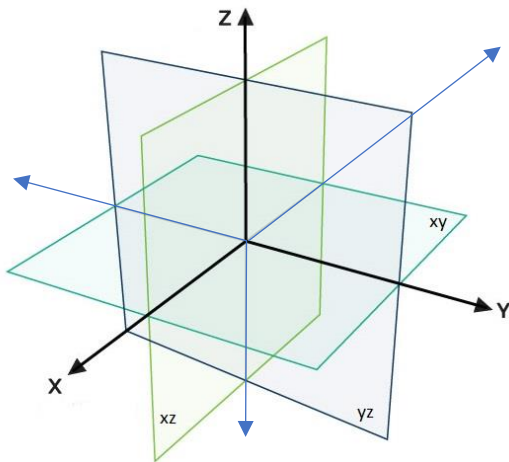
$z = \sqrt[4]{1-y}$, $D = 1 - y \geq 0$, that is $y \leq 1$

$z = \ln \sqrt{x^2 + y^2 - 4}$, $D = x^2 + y^2 - 4 > 0$, $x^2 + y^2 > 2^2$

Acknowledgement: Images are taken from internets. Thanks to owners who have created these beautiful images.

- 1) Graphing Functions
- 2) Level Curves

3D –Coordinates: xyz –Space

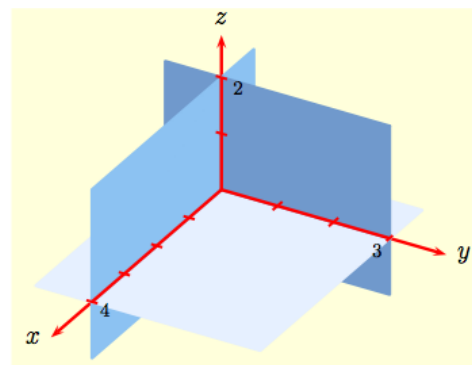
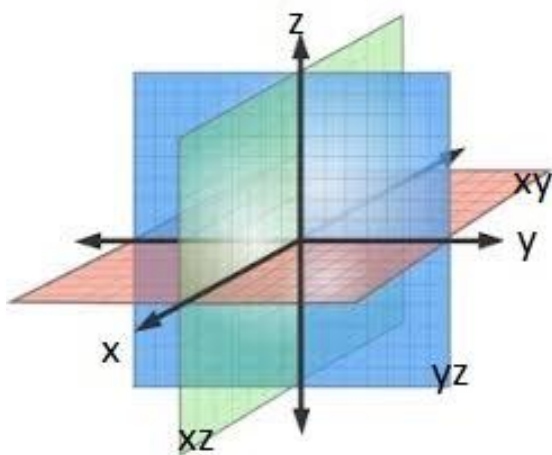
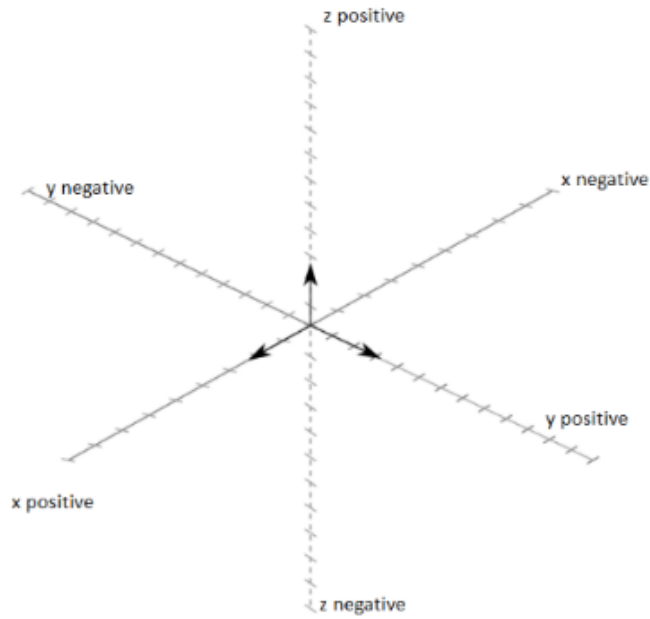


For a point (x, y, z) ,

x is the signed distance from the yz –plane

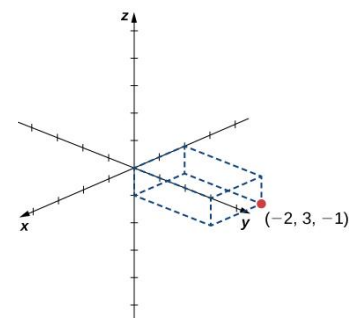
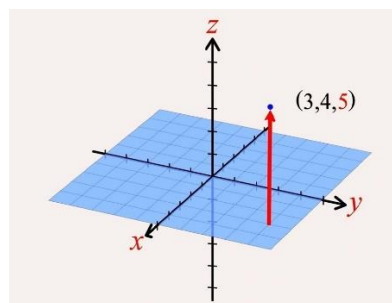
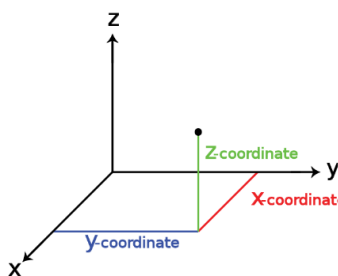
y is the signed distance from the xz –plane

z is the signed distance from the xy –plane



One Viewpoint: $x, y, z > 0$

There are **eight octants** in xyz -space. In the first octant, $x, y, z \geq 0$.



$$(x, y, z) \rightarrow ((x, y), z)$$

Coordinates of the origin $(0, 0, 0)$.

$x = 0$ on the yz –plane. So, the equation of the yz –plane is $x = 0$.

$y = 0$ on the xz –plane. So, the equation of the xz –plane is $y = 0$.

$z = 0$ on the xy –plane. So, the equation of the xy –plane is $z = 0$.

On the x –axis, coordinates of a point is given by $(x, 0, 0)$.

On the y –axis, coordinates of a point is given by $(0, y, 0)$.

On the z –axis, coordinates of a point is given by $(0, 0, z)$.

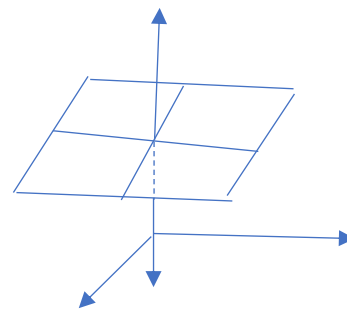
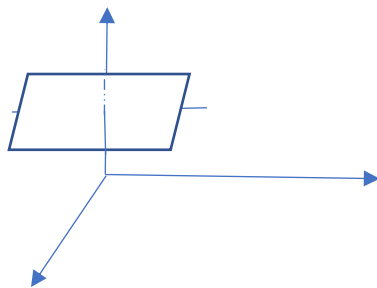
Note: In the 2D, $y = c$, c –constant $\rightarrow$ is a horizontal line, perpendicular to the y –axis at the point $(0, c)$ and parallel to the x –axis.

Graphs of the function $z = f(x, y)$:

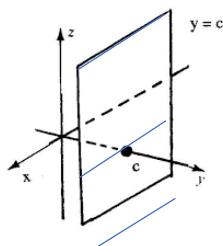
Graphs of Planes:

1) Planes parallel to coordinate planes

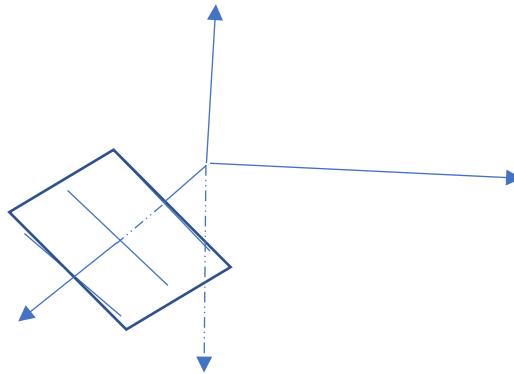
- a. The plane $z = c$ is perpendicular to the z –axis at the point $(0, 0, c)$ and is parallel to the xy –plane.



- b. The plane $y = c$ is perpendicular to the y –axis at the point $(0, c, 0)$ and is parallel to the xz –plane.



- c. The plane $x = c$ is perpendicular to the x -axis at the point $(c, 0, 0)$ and is parallel to the yz -plane.



2) Graph of the plane $x + y + z = 1$.

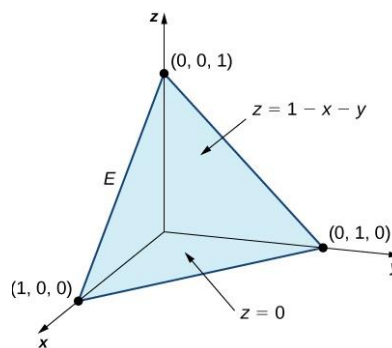
➔ Portion of the plane in the First Octant.

Solution: Given, $x + y + z = 1$. That is, $z = 1 - x - y$, i.e., $z = f(x, y) = 1 - x - y$.

x -intercept: Point on the x -axis, $y = 0, z = 0$. So, the point on the x -axis is given by $(1, 0, 0)$.

y -intercept: Point on the y -axis, $x = 0, z = 0$. So, the point on the y -axis is given by $(0, 1, 0)$.

z -intercept: Point on the z -axis, $x = 0, y = 0$. So, the point on the z -axis is given by $(0, 0, 1)$.



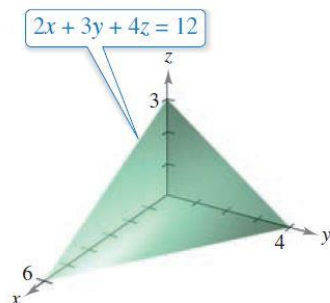
Given, $x + y + z = 1$. That is, $z = 1 - x - y$, that is, $z = f(x, y) = 1 - x - y$.

On the xy -plane: $z = 0$. We get $x + y = 1$.

On the yz -plane: $x = 0$. We get $y + z = 1$.

On the xz -plane: $y = 0$. We get $x + z = 1$.

3) Graph of the plane $2x + 3y + 4z = 12$. Homework



Definition: For any real number x , $\sqrt{x^2} = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$

$\sqrt{x^2} = x$ only for $x \geq 0$. $\sqrt{9} = 3$ and $-\sqrt{9} = -3$.

This means, if the real number y is given by $y = \pm\sqrt{x}$, then by the equation $y = \sqrt{x}$ we get only non-negative real values of y . This is called the positive square root. The negative square root is given by $y = -\sqrt{x}$ which gives only non-positive real values.

$[y = x^2 \text{ is one-to-one } (-\infty, 0], \text{ Inverse function: } y = -\sqrt{x}$

and $y = x^2 \text{ is one-to-one } [0, \infty), \text{ Inverse function: } y = \sqrt{x}$

Note that $\sqrt{4} = \sqrt{(-2)^2} = |\pm 2| = 2$ and $-\sqrt{4} = -2$.

$$y = -|x| = \begin{cases} -(-x), & x < 0 \\ -(x), & x \geq 0 \end{cases} = \begin{cases} x, & x < 0 \\ -x, & x \geq 0 \end{cases}$$

Solve the equation $x^2 = 4 \Rightarrow \sqrt{x^2} = \pm\sqrt{4} = \pm 2$.

$\sqrt{x^2} = |x| = \sqrt{4} \Rightarrow |x| = 2$, that is, $x = \pm\sqrt{4} = \pm 2$.

Note: If $x^2 = 9$, then $x = \pm\sqrt{9} = \pm 3$. Here, $x = \sqrt{9} = 3$, $x = -\sqrt{9} = -3$

4) Graph the function $f(x, y) = \sqrt{x^2 + y^2}$.

Solution: Set $z = f(x, y) = \sqrt{x^2 + y^2}$. So, $z = \sqrt{x^2 + y^2}$. Hence, $z \geq 0$.

On the xy -plane: $z = 0 \Rightarrow \sqrt{x^2 + y^2} = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow x = y = 0$.

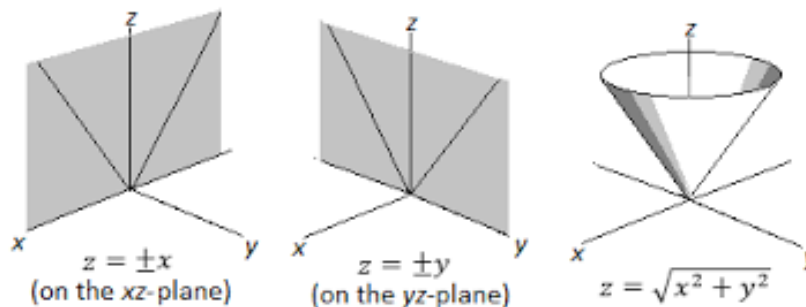
Hence the only point on the xy -plane is $(0, 0, 0)$.

On the yz -plane: $x = 0 \Rightarrow z = \sqrt{0 + y^2} = \sqrt{y^2} \Rightarrow z = |y| = \begin{cases} -y, & y < 0 \\ y, & y \geq 0 \end{cases}$

[Recall: $y = |x|$, V - shaped graph, opening upward on the xy - plane,

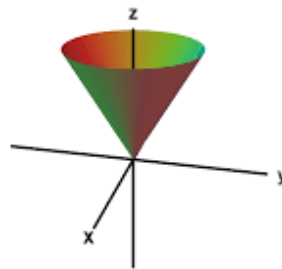
symmetric about the y - axis, y - output, Vertex = $(0, 0)$]

On the xz -plane: $y = 0 \Rightarrow z = \sqrt{x^2 + 0} \Rightarrow z = |x| = \begin{cases} -x, & x < 0 \\ x, & x \geq 0 \end{cases}$



where $z \geq 0$.

The graph of the function $z = f(x, y) = \sqrt{x^2 + y^2}$



Homework:

We know the graph of the function $z = f(x, y) = \sqrt{x^2 + y^2}$

- 5) Graph the function $z = f(x, y) = -\sqrt{x^2 + y^2} \rightarrow$ Reflection about the xy -plane
- 6) Graph the function $z = f(x, y) = 4 - \sqrt{x^2 + y^2} = -\sqrt{x^2 + y^2} + 4 \rightarrow$ Reflection about the xy -plane and shifting 4 units upwards

7) Graph the function $f(x, y) = \sqrt{x^2 + y^2} + 1$. shifting 1 unit upwards

8) Please submit: Graph the function $f(x, y) = -\sqrt{x^2 + y^2} - 1$

Note: If we know the graph of $y = f(x)$, then

(i) we get $y = -f(x)$ by taking the reflection about the x -axis

(ii) we get $y = f(x) + 2$ by shifting the graph upwards by 2 units

we get $y = f(x) - 2$ by shifting the graph downwards by 2 units

$[y = |x|$ and $y = -|x|$ are x -axis reflection of one another.

Similarly, $z = \sqrt{x^2 + y^2}$ and $z = -\sqrt{x^2 + y^2}$ are xy -plane reflection of one another]

9) Graph the function $f(x, y) = x^2 + y^2$.

Solution: Set $z = f(x, y) = x^2 + y^2$. So, $z = x^2 + y^2$. Hence, $z \geq 0$.

On the xy -plane: $z = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow x = y = 0$.

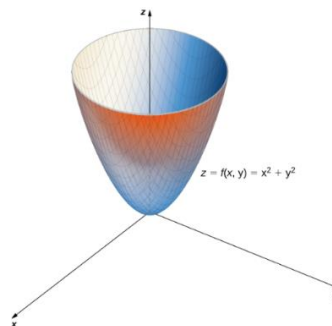
Hence the only point on the xy -plane is $(0, 0, 0)$.

On the yz -plane: $x = 0 \Rightarrow z = y^2$

[Recall: $y = x^2$, U - shaped graph, opening upward on the xy - plane, symmetric about the y - axis, y - output, Vertex = $(0,0)$]

On the xz -plane: $y = 0 \Rightarrow z = x^2$;

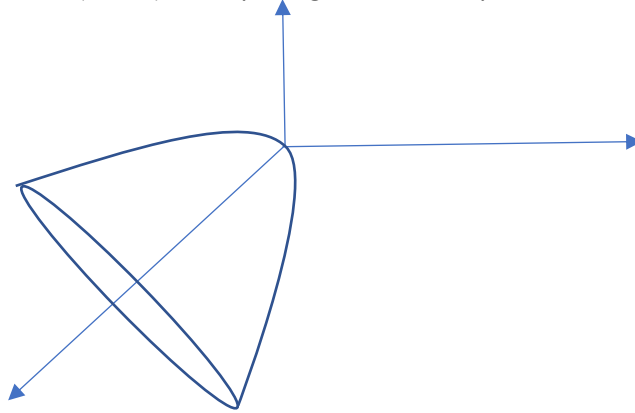
U - shaped graph, opening upward on the xz - plane, symmetric about the z - axis, Vertex = $(0,0)$



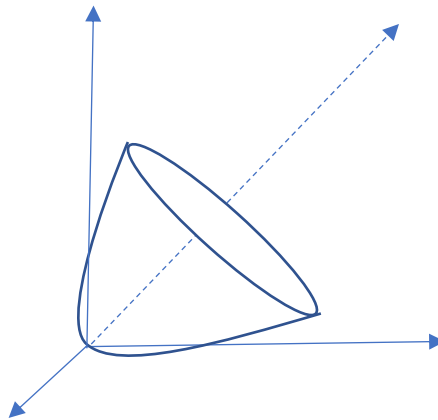
The graph is a paraboloid $z = x^2 + y^2$.

[Exercise 9 is done here]

Note that the graph is a paraboloid $x = f(y, z) = y^2 + z^2$. It's a paraboloid, symmetric about the x -axis, with vertex at the $(0, 0, 0)$ and opening towards the positive x -axis.



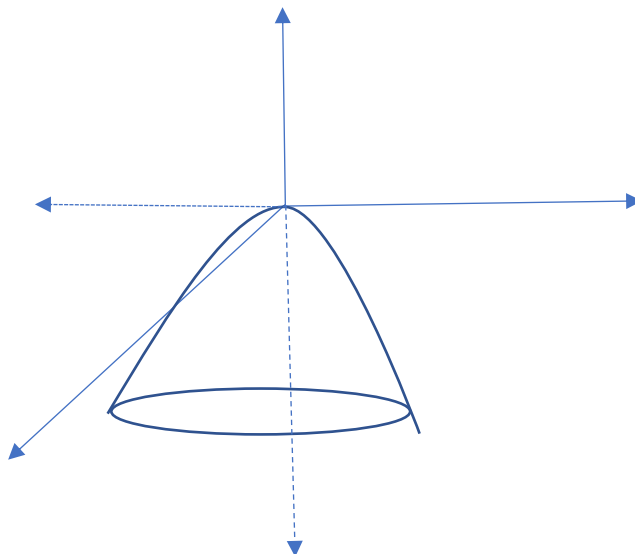
The graph of $x = -(y^2 + z^2)$ is the reflection of the graph of $f(y, z) = y^2 + z^2$ about the yz -plane



Homework:

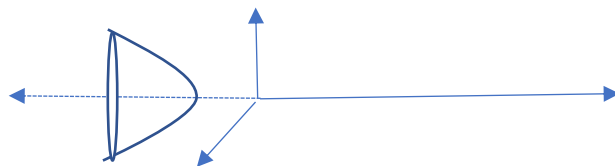
10) Graph the function $z = -(x^2 + y^2)$.

The graph $z = -(x^2 + y^2)$ is the reflection of the graph of $f(x, y) = x^2 + y^2$ about the xy -plane.



If we know the graph of $z = x^2 + y^2$, then

- 11) Graph the function $f(x, y) = 4 - (x^2 + y^2)$.
- 12) Graph the function $f(x, y) = x^2 + y^2 + 1$.
- 13) Graph the function $f(x, y) = -[(x^2 + y^2) + 3]$.
- 14) Graph the function $f(x, z) = -[(x^2 + z^2) + 3]$.



Note that the graph of $z_2 = 4(x^2 + y^2)$ is vertically stretched of the graph $z_1 = x^2 + y^2$.

Sphere: $x^2 + y^2 + z^2 = \rho^2$.

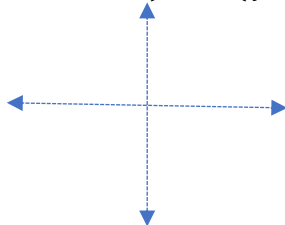
Ellipsoid: $ax^2 + by^2 + cz^2 = \rho^2$, $a, b, c > 0$ and $a \neq b \neq c$.

Q and A Session: 1

1. Sketch the domain of $f(x, y) = \ln(xy)$.

Solution: For the domain of the function, set $xy > 0$, that is,

$$(y > 0 \text{ and } x > 0) \text{ OR } (y < 0 \text{ and } x < 0)$$



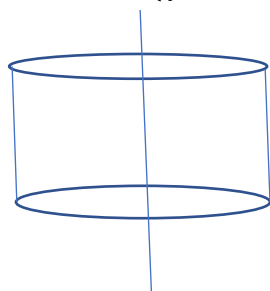
Sketch of D=??

2. Sketch the domain of $f(x, y) = \sqrt{xy}$.

Solution: For the domain of the function, set $xy \geq 0$, that is,

$$(y \geq 0 \text{ and } x \geq 0) \text{ OR } (y \leq 0 \text{ and } x \leq 0)$$

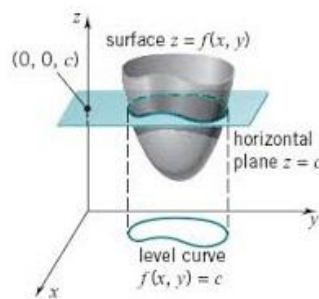
Sketch of D=??



Elliptic Cylinder, base and height of this cylinder is an ellipse.

Definition: Level Curves

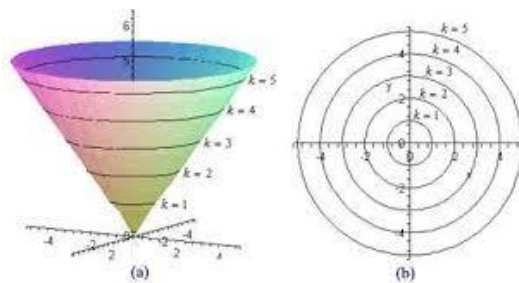
If the surface $z = f(x, y)$ is cut by the horizontal plane $z = k$, then at all the points on the intersection we have $f(x, y) = k$. **The projection of this intersection onto the xy –plane is called the level curve of height k or the level curve with constant k .** A set of level curves of $z = f(x, y)$ is called a contour plot or contour map of f .



For example, for the function $z = f(x, y) = \sqrt{x^2 + y^2}$ the level curves for $z = k$, $k = 0, 1, 2, 3, 4, 5$ are given by the graph

Defined by $f(x, y) = \sqrt{x^2 + y^2} = k$, that is, $\sqrt{x^2 + y^2} = k$, $k = 0, 1, 2, 3, 4, 5$. That is,

$$x^2 + y^2 = k^2, \quad k = 0, 1, 2, 3, 4, 5.$$



Equation $\sqrt{x^2 + y^2} = k$; $k = 0, 1, 2, 3, 4, 5$

$$k = 0: \sqrt{x^2 + y^2} = 0 \Rightarrow x^2 + y^2 = 0 \Rightarrow (0, 0)$$

$$k = 1: \sqrt{x^2 + y^2} = 1 \Rightarrow x^2 + y^2 = 1^2$$

$$k = 5: \sqrt{x^2 + y^2} = 5 \Rightarrow x^2 + y^2 = 5^2$$

Definition: Level Surface

If the surface $w = f(x, y, z)$ is cut by the equation $w = k$, then at all the points on the intersection we have $f(x, y, z) = k$. The projection of this intersection onto the xyz -space is called the level surface with constant k .

For Example, for $w = x^2 + y^2 - z$ with $w = 1$: Level surface: $1 = x^2 + y^2 - z$,

that is, $z = x^2 + y^2 - 1$; so the level surface is a paraboloid opening upward with vertex at the point $(0, 0, -1)$.

13*) Graph the function $f(x, y) = \sqrt{x^2 + y^2 - 1}$. → Please read this, and try to understand.

Set $z = f(x, y) = \sqrt{x^2 + y^2 - 1}$. So, $z = \sqrt{x^2 + y^2 - 1}$. Hence, $z \geq 0$.

Domain: $\{(x, y): x^2 + y^2 \geq 1\}$, Range: $\{z : z \geq 0\}$.

Understanding: Set $z = k$, $k = 0, 1, 2, 3, 6$,

Level curves: $z = \sqrt{x^2 + y^2 - 1}$, $z = k$, $k = 0, 1, 2, 3$,

Now, $\sqrt{x^2 + y^2 - 1} = k$, $k = 0, 1, 2, 3$,

$k = 0$: $x^2 + y^2 = 1$, circle of radius 1 with center at $(0,0)$. Height of the level curve $k = 0$.

$k = 1$: $x^2 + y^2 = 2$, circle of radius $\sqrt{2}$ with center at $(0,0)$. Height of the level curve $k = 1$.

$k = 2$: $x^2 + y^2 = 5$, circle of radius $\sqrt{5}$ with center at $(0,0)$. Height of the level curve $k = 2$.

$k = 3$: $x^2 + y^2 = 10$, circle of radius $\sqrt{10}$ with center at $(0,0)$. Height of the level curve $k = 3$.

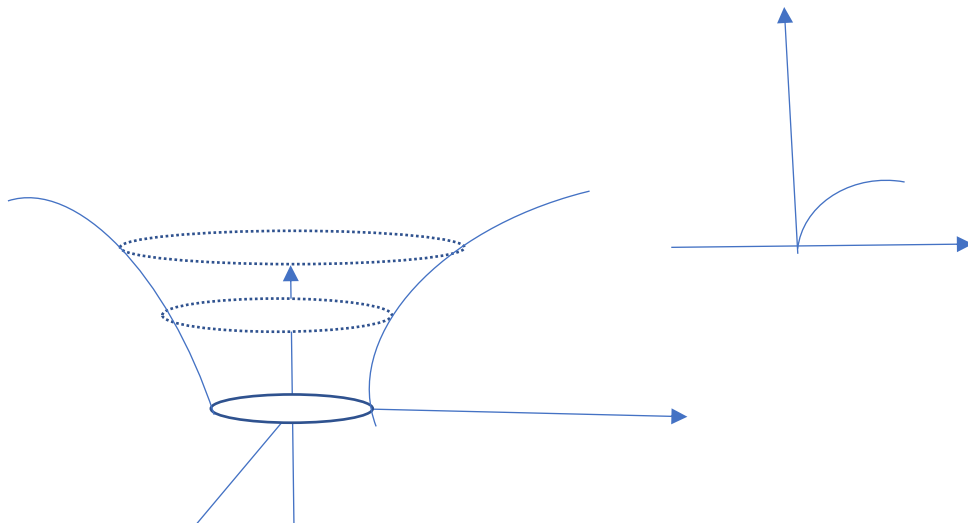
$$z = \sqrt{x^2 + y^2 - 1}$$

Notice that on the xy - plane: $z = 0 \rightarrow 0 = \sqrt{x^2 + y^2 - 1}$, that is, $x^2 + y^2 = 1$.

Notice that on the yz - plane: $x = 0 \rightarrow z = \sqrt{y^2 - 1}$.

Notice that on the xz - plane: $y = 0 \rightarrow z = \sqrt{x^2 - 1}$.

Recall: $y = \sqrt{x}$, y - output axis. This is a graph that is concave downward.



14) Compare the graphs of the function $f(x, y) = \sqrt{x^2 + y^2 - 1}$ and the function

$$f(x, y) = \sqrt{x^2 + y^2 + 1}.$$

$$f(x, y) = \sqrt{x^2 + y^2 + 1} : D = xy\text{-plane}, R = z \geq 1.$$

Hint: You need to understand the level curves to see the graph of $f(x, y) = \sqrt{x^2 + y^2 - 1}$

Section: 13.2 Limits and Continuity

Acknowledgement: All the images are taken from the internet. I cordially thank the owner who have created these images

Theorem: If $y = P(x)$ is a polynomial, then $\lim_{x \rightarrow c} P(x) = P(c)$.

Recall: MAT-120 Calculus – I: Limit $\leftrightarrow$ approximation

Note that, $\lim_{x \rightarrow 1} (x + 1) = 1 + 1 = 2$; $x \neq 1$. Here $x = 0.999999 \dots$, $x = 1.000 \dots 0001$, $x \approx 1$.

Definition in 2D: **limits (an informal view)** If the values of $y = f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (**but not equal to a**), then we write

$$\lim_{x \rightarrow a} f(x) = L$$

which is read “the limit of $f(x)$ as x approaches a is L ” or “ $f(x)$ approaches L as x approaches a .” The expression can also be written as $f(x) \rightarrow L$ as $x \rightarrow a$.

Let’s start with $f(x) = \frac{x^2-1}{x-1}$. Note that $f(1) = \frac{0}{0}$, where $f(x) = x + 1$, $x \neq 1$.

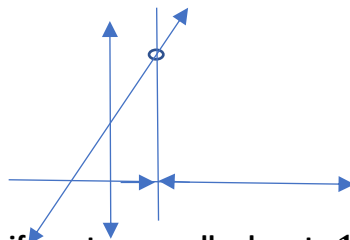
[Note: If $x \neq 0$, then $\frac{x}{x} = 1$. That is, If $x - 1 \neq 0$, then $\frac{x-1}{x-1} = 1$. We know that

$$x + 1 = 1(x + 1) = \frac{x-1}{x-1}(x + 1), \quad x \neq 1.]$$

If $x = 1$, then $y = 1 + 1 = 2$. Since $x \neq 1$, hence $y \neq 2$. So, the graph of the function

$$y = f(x) = x + 1, \quad x \neq 1. \quad D = \text{All real, except } 1$$

is a straight line which does not contain the point $(1, 2)$. That means, there is a whole at the point $(1, 2)$.



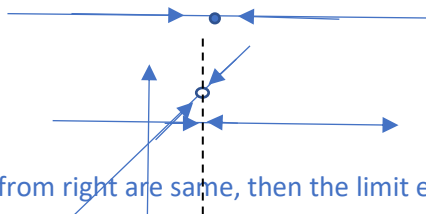
Question: What happen to outputs if inputs are really close to 1? $\lim_{x \rightarrow 1} f(x) = x + 1, \quad x \neq 1$.

x	.99	.999	0.9999	0.999999	$x = 1$	1.000001	1.0001	1.001	1.01
$y = f(x)$	1.99	1.999	1.9999	1.999999	$\frac{0}{0}$	2.000001	2.0001	2.001	2.01

Left limit (approximation) of the function $y = f(x)$ is $1.999999 \approx 2$

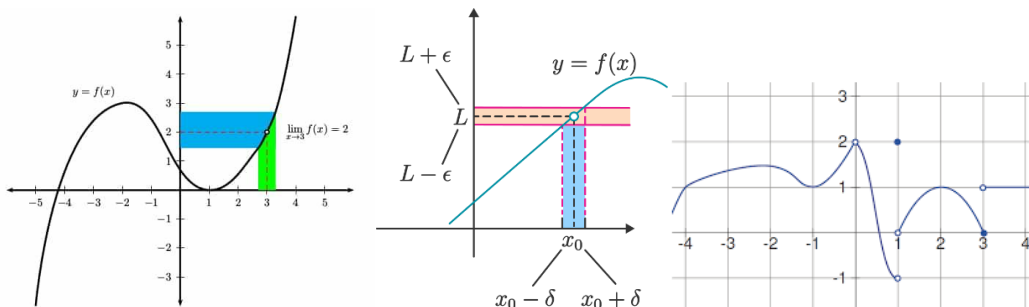
Right limit (approximation) of the function $y = f(x)$ is $2.000001 \approx 2$

$$y = f(x) = x + 1, \quad x \neq 1$$



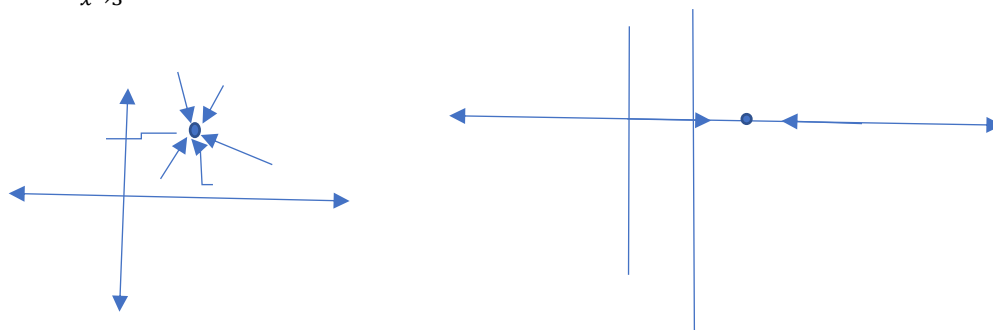
Note: If the limits from left and from right are same, then the limit exists, and the value of the limit is the **common limit**. Here, if $x \neq 1$, but $x \rightarrow 1$, then the limit of $f(x)$ is 2.

Graphical Understanding:



$\lim_{x \rightarrow 0} f(x) = 2$, $\lim_{x \rightarrow 1^-} f(x) = -1$ and $\lim_{x \rightarrow 1^+} f(x) = 0$, that is $\lim_{x \rightarrow 1} f(x)$ does not exist.

Here, $\lim_{x \rightarrow 3} f(x)$ does not exist.



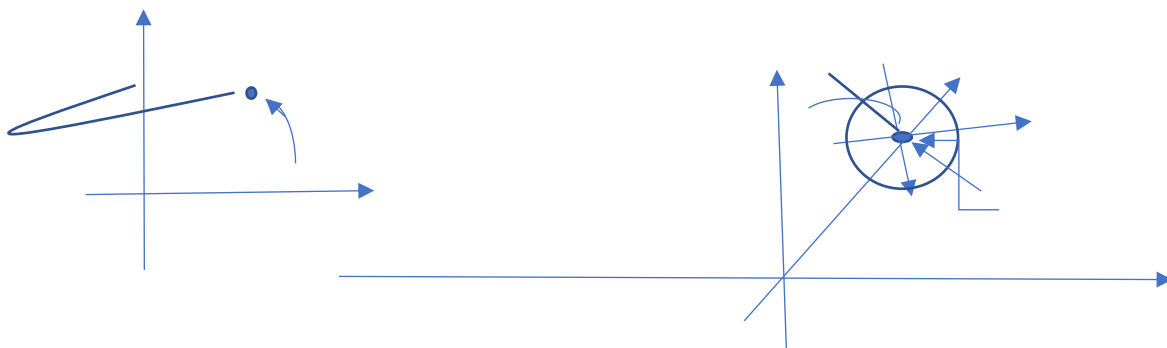
Alternative Limit Definition

Let $f(x)$ be defined for all x in some open interval containing the number x_0 , with the possible exception that $f(x)$ need not be defined at x_0 . We will write $\lim_{x \rightarrow x_0} f(x) = L$

if given any number $\varepsilon > 0$ we can find a number $\delta > 0$ such that

$$|f(x) - L| < \varepsilon \quad \text{if} \quad 0 < |x - x_0| < \delta.$$

Lecture 5: MAT-250 $z = f(x, y)$, Domain $\rightarrow$ 2 Dimensional



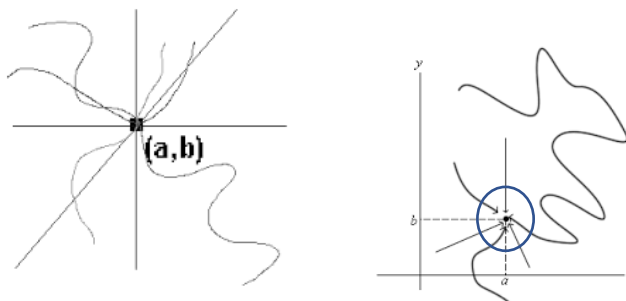
$$(x - x_0)^2 + (y - y_0)^2 = r^2 ; \text{ If } r \rightarrow 0, \text{ then } (x, y) \rightarrow (x_0, y_0)$$

$y = f(x)$ ← function of single variable. Limit: 1.1, 1.2, 1.3, 1.6 and 3.6.

$z = f(x, y)$ ← function of two variables. A direction=a smooth curve=an equation $y = g(x)$

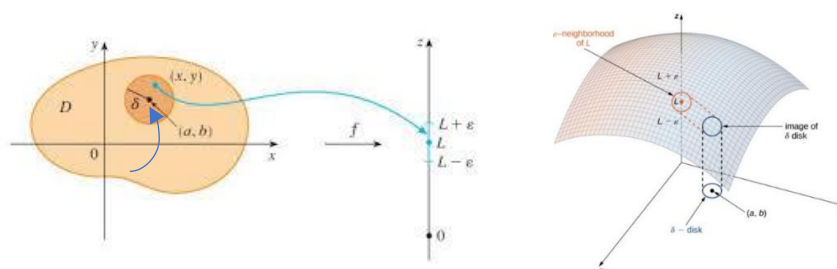
In 3D: Function of Two Variables $z = f(x, y)$

The domain is 2-dimensional. And, there are infinitely many directions to approach a point on the plane.



$$z = f(x, y)$$

$$3D \text{ Graph} = \{(x, y, z) | z = f(x, y)\}$$



$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$$

Definition:

Let f be a function of two variables, and assume that f is defined at all points of some open disk centered at (a, b) , except possibly at (a, b) . We will write

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$$

if given any number $\varepsilon > 0$, we can find a number $\delta > 0$ such that $f(x,y)$ satisfies

$$|f(x,y) - L| < \varepsilon$$

whenever the distance between (x,y) and (a,b) satisfies

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta.$$

$[\varepsilon, \delta > 0, \text{ and } \varepsilon, \delta \rightarrow 0]$

Theorem

- a) If $f(x,y) \rightarrow L$ as $(x,y) \rightarrow (a,b)$, then $f(x,y) \rightarrow L$ as $(x,y) \rightarrow (a,b)$ along any smooth curve.
- (b) If the limit of $f(x,y)$ fails to exist as $(x,y) \rightarrow (a,b)$ along some smooth curve, or if $f(x,y)$ has different limits as $(x,y) \rightarrow (a,b)$ along two different smooth curves, then the limit of $f(x,y)$ does not exist as $(x,y) \rightarrow (a,b)$.

Note: Since we are comfortable to work with a function of single variable, so we can use the equation of the smooth curve to replace one variable from the function.

Note: If $(x,y) \rightarrow (0,0)$, then x and y are **not simultaneously zero**. Also, if $(x,y) \rightarrow (0,0)$, then $x \rightarrow 0$ and $y \rightarrow 0$.

Understanding:

$$(x,y) \rightarrow (0,0) \Rightarrow (x,y) \neq (0,0)$$

$$(x,y) \rightarrow (0,0) \text{ along the } y\text{-axis} \Rightarrow (0,y) \rightarrow (0,0), y \rightarrow 0.$$

$$(x,y) \rightarrow (0,0) \text{ along the } x\text{-axis} \Rightarrow (x,0) \rightarrow (0,0), x \rightarrow 0.$$

Understanding: If $x \rightarrow -1$, then $x^2 - 3 \rightarrow -2$ and $x^2 + 1 \rightarrow 2$. Also, If $x \rightarrow 0$, then $x^2 \rightarrow 0$.

Q and A session

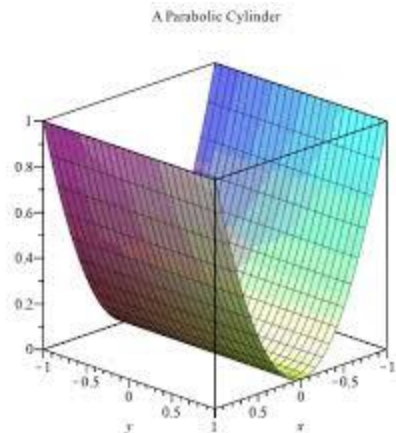
Q1) 13.1

Graph $f(x,y) = x^2$ ← function of two variable, y is a free variable.

[$x = 1$ is a line. Graph = $\{(x,y) : x = 1 \text{ and } y \text{ is a free variable}\}$]

$$z = f(x,y) = x^2.$$

It's a parabola on the xz -plane and y is a free variable.



Goal: We will try to write the function as a function of single variable. [1.1, 1.2, 1.3, 1.6 and 3.6]

Understanding:

If $(x, y) \rightarrow (a, b)$ and $y = f(x)$ is the smooth curve that goes through the point (a, b) , then $x \rightarrow a$ implies $y \rightarrow f(a) = b$. Note that, we have to choose $y = f(x)$ so that (a, b) is on the graph of $y = f(x)$.

Example: 1A Show that the limit $\lim_{(x,y) \rightarrow (0,0)} -\frac{yx}{x^2+y^2}$ does not exist.

Direction = Path = curve = equation!!!

Solution: Given limit: $\lim_{(x,y) \rightarrow (0,0)} -\frac{yx}{x^2+y^2}$

[Curves: $y = \pm x$; $y = \pm mx$; $y = x^2$; $y = x^n, n > 0$; $y = \sin x$; $y = \sqrt{x}$;]

Note: $\lim_{x \rightarrow 0} k = k$. If $(x, x^2) \rightarrow (0,0)$, then $x \rightarrow 0$.

Along the x-axis: Along $y = 0$. Since $(x, y) \rightarrow (0,0)$ and $y = 0$, hence $(x, 0) \rightarrow (0,0)$. Here $x \rightarrow 0$.

Now, $\lim_{(x,y) \rightarrow (0,0)} -\frac{yx}{x^2+y^2} = \lim_{x \rightarrow 0} -\frac{0(x)}{x^2+0^2} = \lim_{x \rightarrow 0} \left(-\frac{0}{x^2}\right) = \lim_{x \rightarrow 0} (0) = 0$. [$x \neq 0$]

[Note that limits along the y-axis and along the curve $y = x^2$ is also 0] ← Homework

Along the line $y = x$. Since $(x, y) \rightarrow (0,0)$ and $y = x$, hence $(x, x) \rightarrow (0,0)$. Here $x \rightarrow 0$.

Also, $\lim_{(x,y) \rightarrow (0,0)} -\frac{yx}{x^2+y^2} = \lim_{x \rightarrow 0} -\frac{x(x)}{x^2+x^2} = \lim_{x \rightarrow 0} -\frac{x^2}{2x^2} = \lim_{x \rightarrow 0} \left(-\frac{1}{2}\right) = -\frac{1}{2}$; [Since $x^2 \neq 0 \rightarrow \frac{x^2}{x^2} = 1$]

We get two different limits along two different smooth curves $y = 0$ and $y = x$. Hence, the limit does not exist.

[Shown]

How to find limits: [MAT 120]

Substitute the value in the function.

- i) If it gives a finite value, then we are done
- ii) If we get $\frac{0}{\text{nonzero}}$, then the limit is 0 and we are done
- iii) If we get $\frac{0}{0}$, use approximation [Find left and right limit, if necessary]

Exercise: 1B

Show that the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{3y^2x}{x^3-2y^3}$ does not exist.

Solution: Given

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3y^2x}{x^3-2y^3}$$

Note: The function $f(x, y) = \frac{3y^2x}{x^3-2y^3}$ is indeterminate at $(0,0)$.

- (i) Along the curve x –axis: Equation of the curve $y = 0$.

$(x, y) \rightarrow (0, 0)$ along the x –axis implies $(x, 0) \rightarrow (0, 0)$, i. e., $x \rightarrow 0$. Hence, we get

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3y^2x}{x^3-2y^3} = \lim_{x \rightarrow 0} \frac{3(0)x}{x^3-0} = \lim_{x \rightarrow 0} \frac{0}{x^3}; x \neq 0$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3y^2x}{x^3-2y^3} = 0 \text{ along the smooth curve } y = 0.$$

- (ii) Along the curve $y = x$:

$(x, y) \rightarrow (0, 0)$ along the curve $y = x$ implies $(x, x) \rightarrow (0, 0)$, i. e., $x \rightarrow 0$. Hence, we get

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3y^2x}{x^3-2y^3} = \lim_{x \rightarrow 0} \frac{3x^2x}{x^3-2x^3} = \lim_{x \rightarrow 0} \frac{3x^3}{-x^3} = \lim_{x \rightarrow 0} (-3); x \neq 0$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3y^2x}{x^3 - 2y^3} = -3 \text{ along the smooth curve } y = x.$$

Hence $\lim_{(x,y) \rightarrow (0,0)} \frac{3y^2x}{x^3 - 2y^3}$ has different value along the two smooth curves, therefore, the limit does not exist.

Showed.

Exercise: 1C Evaluate the limit $\lim_{(x,y) \rightarrow (1,-2)} \frac{3yx}{x^2 + 2y^2} = \frac{3(-2)(1)}{1^2 + 2(-2)^2} = -\frac{6}{9} = -\frac{2}{3}.$

Definition: The function $f(x, y)$ is continuous at the point (a, b) if and only if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

Note that the function $f(x, y) = \frac{3yx}{x^2 + 2y^2}$ is continuous at the point $(1, -2)$.

Exercise: 1D Evaluate the limit $\lim_{(x,y) \rightarrow (1,-1)} \frac{3yx}{x^2 - y^2}$; by substitution, we get $-\frac{3}{0}$

Remember that $z = \frac{3yx}{x^2 - y^2}$

Consider the smooth curve $y = -1$.

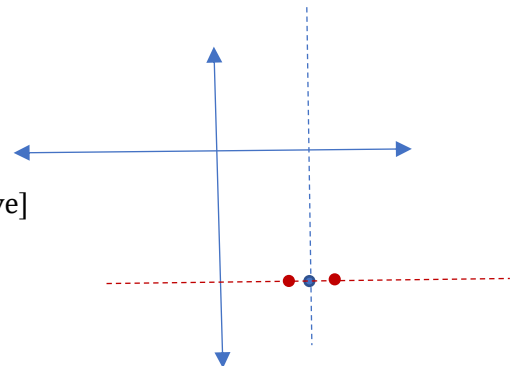
So, $(x, y) \rightarrow (1, -1)$ along the curve $y = -1$, implies

$(x, -1) \rightarrow (1, -1)$, i. e., $x \rightarrow 1$ [x is close to 1 and hence positive]

Now, we get

$$\lim_{(x,y) \rightarrow (1,-1)} \frac{3yx}{x^2 - y^2} = \lim_{x \rightarrow 1} \frac{-3x}{x^2 - 1} = \begin{cases} \lim_{x \rightarrow 1^-} \frac{-3x}{x^2 - 1} = \infty \\ \lim_{x \rightarrow 1^+} \frac{-3x}{x^2 - 1} = -\infty \end{cases}$$

Limit does not exist.



Example: 2 Find this limit $\lim_{(x,y) \rightarrow (0,0)} \frac{3yx}{x^2+2y^2}$ along the following curves:

(a) the x –axis (b) the y –axis (c) the line $y = x$.

(d) the line $y = -x$ (e) the parabola $y = x^2$.

Solution: Part (e)

Along the line $y = x^2$. Since $(x, y) \rightarrow (0,0)$ and $y = x^2$, hence $(x, x^2) \rightarrow (0, 0)$. Here $x \rightarrow 0$.

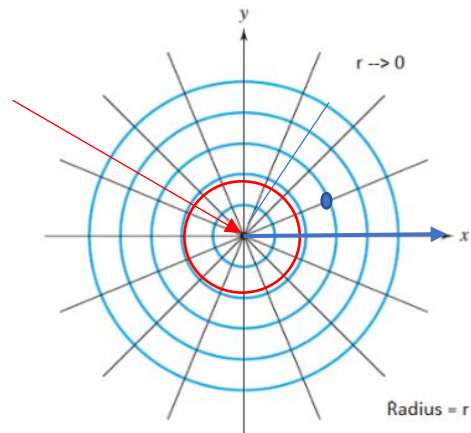
$$\lim_{(x,y) \rightarrow (0,0)} \frac{3yx}{x^2 + 2y^2} = \lim_{x \rightarrow 0} \frac{3(x^2)x}{x^2 + 2(x^2)^2} = \lim_{x \rightarrow 0} \frac{3x^3}{x^2 + 2x^4} = \lim_{x \rightarrow 0} \frac{3x^3}{x^2(1 + 2x^2)} ; x^2 \neq 0.$$

$$= \lim_{x \rightarrow 0} \frac{3x}{1 + 2x^2} = \frac{0}{1} = 0$$

Exercise: 3 Show that the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{3yx}{x^2+2y^2}$ does not exist.

Exercise: 4 Evaluate the limit $\lim_{(x,y) \rightarrow (1,-2)} \frac{3yx}{x^2+2y^2} = \frac{3(-2)1}{1^2+2(-2)^2} = -\frac{6}{9} = -\frac{2}{3}$.

Understanding:



MAT 130: Polar Coordinates $(r, \theta) \rightarrow \theta \in \mathbb{R}$; Equation of the polar axis: $\theta = 0$. To form a circle θ must be $0 \leq \theta \leq 2\pi$. Now, if $r \rightarrow 0^+$, $r > 0$ gives an approximation that is a point circle at the origin.

Theorem 1 (L'Hôpital's Rule for Form $\frac{0}{0}$)

Suppose that f and g are differentiable functions on an open interval containing $x = a$, except possibly at $x = a$, and that $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$.

If $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists, or if this limit is ∞ or $-\infty$, then $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{0}{0}$ form, and then by the rule

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}.$$

Moreover, this statement is also true in the case of a limit as $x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow -\infty$, or as $x \rightarrow +\infty$.

Theorem 2 (L'Hôpital's Rule for Form $\frac{\infty}{\infty}$)

Suppose that f and g are differentiable functions on an open interval containing $x = a$, except possibly at $x = a$, and that $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$.

If $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists, or if this limit is ∞ or $-\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Moreover, this statement is also true in the case of a limit as $x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow -\infty$, or as $x \rightarrow +\infty$.

Note: $0 \cdot \infty \rightarrow \frac{1}{\infty} \cdot \infty \rightarrow \frac{\infty}{\infty}$ OR $0 \cdot \infty \rightarrow 0 \cdot \frac{1}{0} \rightarrow \frac{0}{0}$.

Goal: Convert the function to a function of single variable. (Cal-I: MAT-120 $\rightarrow$ 1.1, 1.2, 1.3, 1.6 and 3.6)

Example: 4 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} e^{-\frac{1}{x^2+y^2}}$.

Solution: Set $z = x^2 + y^2$. Since $(x, y) \rightarrow (0, 0)$ and $z = x^2 + y^2$, then $z = x^2 + y^2 > 0$ and $z \rightarrow 0^+$.

Here, we have $(\sqrt{z})^2 = x^2 + y^2$: a circle of radius $\sqrt{z}$, $\sqrt{z} \rightarrow 0$.

So,

$$\lim_{(x,y) \rightarrow (0,0)} e^{-\frac{1}{x^2+y^2}} = \lim_{z \rightarrow 0^+} e^{-\frac{1}{z}} = \lim_{z \rightarrow 0^+} \frac{1}{e^{\frac{1}{z}}} ; \left[e > 2 \text{ and } \lim_{u \rightarrow \infty} e^u = \infty \ \& \ \lim_{v \rightarrow 0^+} \frac{1}{v} = \infty \right]$$

$$= \frac{1}{\lim_{z \rightarrow 0^+} e^{\frac{1}{z}}} = 0$$

[Note: Notice that $x^2 + y^2 = (\sqrt{z})^2$. Think: $\frac{1}{0} = \infty$, $\frac{1}{\infty} = 0$, $2^\infty \rightarrow \infty$, $\left(\frac{1}{2}\right)^\infty = 2^{-\infty} \rightarrow 0$]

$$\lim_{(x,y) \rightarrow (0,0)} e^{-\frac{1}{x^2+y^2}} = \lim_{z \rightarrow 0^+} e^{-\frac{1}{z}} = \lim_{z \rightarrow 0^+} \frac{1}{e^{\frac{1}{z}}} = \frac{1}{\lim_{z \rightarrow 0^+} e^{\frac{1}{z}}} = 0;$$

; since $\lim_{x \rightarrow 0^+} \frac{1}{x} = \infty$, $\lim_{x \rightarrow \infty} e^x = \infty$ and $\lim_{x \rightarrow \infty} \frac{1}{x} = 0$.

[1.6: Theorem: If f is continuous, then $\lim_{z \rightarrow a} f(g(z)) = f\left(\lim_{z \rightarrow a} g(z)\right)$.

Here $f(x) = e^x$, $g(x) = \frac{1}{x}$.]

Example: 5 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{-\frac{1}{\sqrt{x^2+y^2}}}}{\sqrt{x^2+y^2}}$.

Solution: Set $z = \sqrt{x^2 + y^2}$. Since $(x, y) \rightarrow (0, 0)$, then $z \rightarrow 0^+$.

$$\lim_{(x,y) \rightarrow (0,0)} \frac{e^{-\frac{1}{\sqrt{x^2+y^2}}}}{\sqrt{x^2+y^2}} = \lim_{z \rightarrow 0^+} \frac{e^{-\frac{1}{z}}}{z} ; \left[\frac{\lim_{z \rightarrow 0^+} e^{-\frac{1}{z}}}{\lim_{z \rightarrow 0^+} z} \text{ is in the form } \frac{0}{0} \right]$$

[Note: $\frac{d}{dx} \left(e^{-\frac{1}{z}} \right) = \frac{1}{z^2} e^{-\frac{1}{z}}$] $[e^{-\frac{1}{0}} = e^{-\infty} = \frac{1}{e^\infty} = \frac{1}{\infty} = 0; \frac{0}{0} = \frac{\frac{1}{\infty}}{\frac{1}{\infty}} = \frac{\infty}{\infty} = \frac{\frac{1}{0}}{\frac{1}{0}} = \frac{0}{0}]$

$$= \lim_{z \rightarrow 0^+} \frac{1}{ze^{\frac{1}{z}}} ; [0 \cdot \infty \text{ form}]$$

$$= \lim_{z \rightarrow 0^+} \frac{\frac{1}{z}}{\frac{1}{e^{\frac{1}{z}}}} \text{ is in the form } \frac{\infty}{\infty} ; \left[\frac{\frac{1}{a}}{b} = \frac{1}{ab} = \frac{\frac{1}{b}}{a} \right].$$

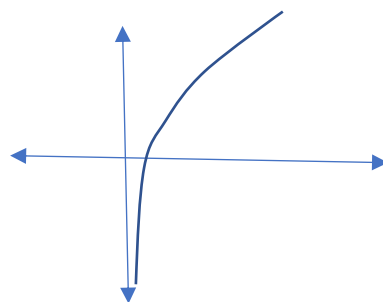
$$= \lim_{z \rightarrow 0^+} \frac{\frac{1}{z^2}}{\frac{1}{e^{\frac{1}{z}}} \cdot \frac{-1}{z^2}} ; \text{Applying L'Hopital's Rule}$$

$$= \lim_{z \rightarrow 0^+} \frac{1}{e^{\frac{1}{z}}} = 0.$$

[Alternative Method: set $w = \frac{1}{z}$. Then $z = \frac{1}{w}$. Also, if $z \rightarrow 0^+$, then $w \rightarrow \infty$.

We get,

$$\begin{aligned} & \lim_{z \rightarrow 0^+} \frac{\frac{1}{z}}{e^{\frac{1}{z}}} \\ &= \lim_{w \rightarrow \infty} \frac{w}{e^w} \quad [\text{This is in } \frac{\infty}{\infty}, \text{ so we can use L'Hopital's Rule}] \\ &= \lim_{w \rightarrow \infty} \frac{1}{e^w} = 0. \end{aligned}$$



3.6 MAT-120: $\lim_{z \rightarrow 0^+} \tan z \ln z = \lim_{z \rightarrow 0^+} \frac{\ln z}{\cot z}$ and $\lim_{z \rightarrow 0^+} \ln z = -\infty$.

Formula (1.6): (1) $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1 = \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = \lim_{x \rightarrow 0^-} \frac{\sin x}{x}$, that is, $\lim_{* \rightarrow 0} \frac{\sin *}{*} = 1$.

(2) $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} = 0$, that is, $\lim_{\# \rightarrow 0} \frac{1 - \cos \#}{\#} = 0$

(3) $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$

Exercise: 6

$$\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}}$$

Solution: Set $p = \sqrt{x^2 + y^2 + z^2}$. Since $(x, y, z) \rightarrow (0, 0, 0)$, so $p \rightarrow 0^+$. Then

$$\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{\sin(\sqrt{x^2 + y^2 + z^2})}{\sqrt{x^2 + y^2 + z^2}} = \lim_{p \rightarrow 0^+} \frac{\sin p}{p} = 1.$$

Or by L'Hopital's $\lim_{p \rightarrow 0^+} \frac{\sin p}{p} = \lim_{p \rightarrow 0^+} \frac{\cos p}{1} = \cos 0 = 1$.

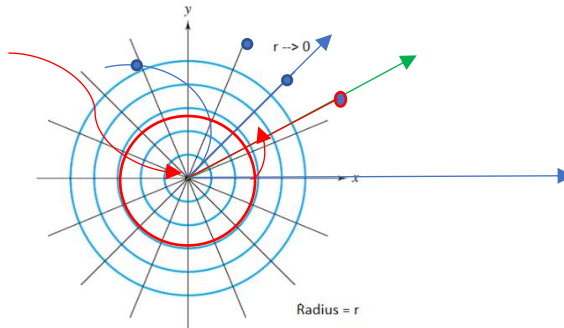
Homework: Practice Exercises from the list given.

Polar Coordinate: For (x, y) and (r, θ) : $x = r \cos \theta$, $y = r \sin \theta$

$$r^2 = x^2 + y^2, \text{ that is } r = \sqrt{x^2 + y^2}.$$

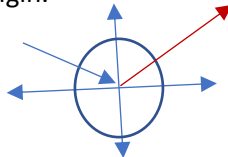
Note that $r \geq 0$ and $r = 0$ only at the origin.

In polar coordinate, the equation of a circle with center at the origin and radius $k > 0$ is given by $r = k$.



MAT 130: Polar Coordinates $(r, \theta) \rightarrow \theta \in \mathbb{R}$; Equation of the polar axis: $\theta = 0$. To form a circle θ must be $0 \leq \theta \leq 2\pi$. Now, if $r \rightarrow 0^+$, $r > 0$ gives an approximation that is a point circle at the origin.

Note: If $x = r \cos \theta$, $y = r \sin \theta$, and if $r \rightarrow 0$, then by this we understand the set of all point on some small disk around the origin, but not the origin.



Example: 6 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2)^3$.

Solution: Set $x = r \cos \theta$, $y = r \sin \theta$. Then

$$r^2 = x^2 + y^2, \text{ that is } r = \sqrt{x^2 + y^2}.$$

Since $(x, y) \rightarrow (0, 0)$ and $r = \sqrt{x^2 + y^2}$, implies $r \rightarrow 0^+$. Now, we have

$$\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2)^3 = \lim_{r \rightarrow 0^+} r^2 \ln(r^2)^3 \quad ; \quad 0 \leq \theta \leq 2\pi.$$

$$= \lim_{r \rightarrow 0^+} r^2 \ln r^6$$

$$= 6 \cdot \lim_{r \rightarrow 0^+} r^2 \ln r \quad ; \text{ this is in the form } 0 \cdot \infty$$

$$= 6 \cdot \lim_{r \rightarrow 0^+} \frac{\ln r}{\frac{1}{r^2}} \quad ; \text{ this is in the form } \frac{\infty}{\infty}, \quad \text{hence apply L'Hopital's Rule}$$

$$[\text{Note: } r \rightarrow 0^+ : r^2 \rightarrow 0^+ \text{ \& } \ln r \rightarrow -\infty] \quad \left[0 = \frac{1}{\frac{1}{0}} = \frac{1}{\infty} \quad \& \quad \infty = \frac{1}{\frac{1}{\infty}} = \frac{1}{0} \rightarrow r^2 = \frac{1}{\frac{1}{r^2}} \right]$$

$$= 6 \cdot \lim_{r \rightarrow 0^+} \frac{\frac{d}{dr}(\ln r)}{\frac{d}{dr}\left(\frac{1}{r^2}\right)} = 6 \cdot \lim_{r \rightarrow 0^+} \frac{\frac{1}{r}}{-\frac{2}{r^3}} \quad ; \quad \left[\frac{d}{dr}(r^{-2}) = -2r^{-3} \right]$$

$$= 6 \cdot \lim_{r \rightarrow 0^+} \frac{1}{r} \left(-\frac{r^3}{2} \right) \quad ; \quad \left[\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \cdot \frac{d}{c} \right]$$

$$= 6 \cdot \lim_{r \rightarrow 0^+} \left(-\frac{1}{2} r^2 \right) = 6(0) = 0$$

Example: 7 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} y \ln(x^2 + y^2)^3$

Solution:

Using Polar Coordinates, set $x = r \cos \theta$, $y = r \sin \theta$. Then $r = \sqrt{x^2 + y^2} \Rightarrow r^2 = x^2 + y^2$

Since $(x, y) \rightarrow (0, 0)$ and $r = \sqrt{x^2 + y^2}$, so $r \rightarrow 0^+$.

Now, $\lim_{(x,y) \rightarrow (0,0)} y \ln(x^2 + y^2)^3$

$$= \lim_{r \rightarrow 0^+} (r \sin \theta) \ln(r^2)^3 ; \quad 0 \leq \theta \leq 2\pi.$$

$$= \lim_{r \rightarrow 0^+} r \sin \theta \ln r^6 ; \quad -1 \leq \sin \theta \leq 1 \text{ for } 0 \leq \theta \leq 2\pi.$$

$$= \lim_{r \rightarrow 0^+} 6 r \sin \theta \ln r$$

$$= 6 \sin \theta \cdot \lim_{r \rightarrow 0^+} r \ln r ; \text{ this is in } 0 \cdot \infty.$$

$$\left[\text{Note: } 0 \cdot \infty = 0 \cdot \frac{1}{0} = \frac{0}{0} \quad \text{OR} \quad 0 \cdot \infty = \frac{1}{\infty} \cdot \infty = \frac{\infty}{\infty} \right]$$

$$[\text{Note: } 2 = \frac{1}{\frac{1}{2}} ; \quad \ln r = \frac{1}{\frac{1}{\ln r}} = \frac{1}{(\ln r)^{-1}} , \quad r = \frac{1}{\frac{1}{r}} \text{ and For } b > 1, \lim_{x \rightarrow 0^+} \log_b x = -\infty]$$

$$= 6 \sin \theta \lim_{r \rightarrow 0^+} \frac{\ln r}{\frac{1}{r}} ; \text{ this is in } \frac{\infty}{\infty}, \text{ so we can apply L'Hopitals Rule.}$$

$$= 6 \sin \theta \lim_{r \rightarrow 0^+} \frac{\frac{d}{dr}(\ln r)}{\frac{d}{dr}\left(\frac{1}{r}\right)}$$

$$= 6 \sin \theta \lim_{r \rightarrow 0^+} \frac{\frac{1}{r}}{-\frac{1}{r^2}}$$

$$= 6 \sin \theta \lim_{r \rightarrow 0^+} \left(\frac{1}{r} \right) \times \left(-\frac{r^2}{1} \right)$$

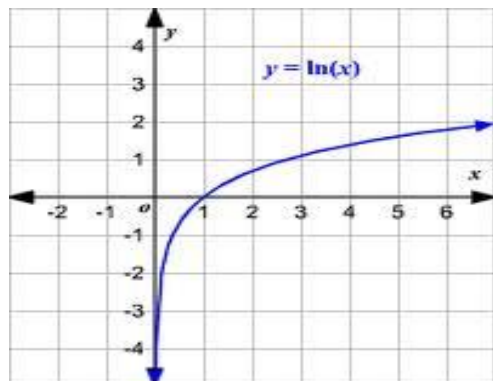
$$= 6 \sin \theta \lim_{r \rightarrow 0^+} (-r) = (6 \sin \theta) \cdot 0 ; \quad \text{where } -1 \leq \sin \theta \leq 1 \text{ for } 0 \leq \theta \leq 2\pi.$$

$$= 0.$$

$$\lim_{(x,y) \rightarrow (0,0)} y \ln(x^2 + y^2)^3 = 0.$$

Note: $\lim_{x \rightarrow q} C f(x) = C \lim_{x \rightarrow q} f(x)$

Understanding: $y = \log_b x$; $b > 1$, increasing function. Note that $y = \ln x$; here base $b = e > 1$.



The y -axis is the vertical asymptote of the function.

Domain: Positive real number = $(0, \infty)$

Range: All real numbers = $(-\infty, \infty)$

y -axis is the vertical Asymptote. The Graph of $y = \ln x$; base $b = e > 1$, will get really close to y -axis, but it will never touch or cross the y -axis.

If $x \rightarrow 0^+$, then $y = \ln x \rightarrow -\infty$, **that is,** $\lim_{x \rightarrow 0^+} \ln x = -\infty$.

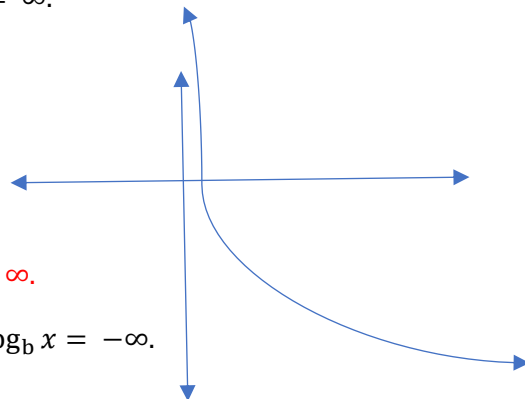
Also, if $x \rightarrow \infty$, then $y = \ln x \rightarrow \infty$, **that is,** $\lim_{x \rightarrow \infty} \ln x = \infty$.

For the logarithmic function of base $0 < b < 1$:

$y = \log_b x$; $0 < b < 1$, decreasing function

If $x \rightarrow 0^+$, then $y = \log_b x \rightarrow \infty$, **that is,** $\lim_{x \rightarrow 0^+} \log_b x = \infty$.

Also, if $x \rightarrow \infty$, then $y = \log_b x \rightarrow -\infty$, **that is,** $\lim_{x \rightarrow \infty} \log_b x = -\infty$.



Example: 8 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} (x^2 + y^2)^3 \ln(x^2 + y^2)^5$.

Homework $\rightarrow$ Similar to Example 6

Example: 9 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} 2xy \ln(x^2 + y^2)$.

Class performance

Example: 10 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - 16y^4}{x^2 + 4y^2} = 0$. [similar to the note below] Homework

Note: For $f(x) = \frac{x^2-1}{x+1} = x-1$, $x \neq -1$; $\lim_{x \rightarrow -1} f(x) = 1-1 = 0$.

Example: 11 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} x^2 y \ln(x^2 + y^2)^2$.

Solution: Using Polar Coordinates, set $x = r \cos \theta$, $y = r \sin \theta$. Then $r = \sqrt{x^2 + y^2}$

$\Rightarrow r^2 = x^2 + y^2$. Since $(x, y) \rightarrow (0,0)$, so $r \rightarrow 0^+$.

$$\lim_{(x,y) \rightarrow (0,0)} x^2 y \ln(x^2 + y^2)^2$$

$$= \lim_{r \rightarrow 0^+} (r^2 \cos^2 \theta)(r \sin \theta) \ln(r^2)^2$$

$$= \cos^2 \theta \sin \theta \lim_{r \rightarrow 0^+} r^3 \ln r^4 ; -1 \leq \sin \theta \leq 1 \text{ and } 0 \leq \cos^2 \theta \leq 1 \text{ for } 0 \leq \theta \leq 2\pi.$$

$$= 4 \cos^2 \theta \sin \theta \lim_{r \rightarrow 0^+} r^3 \ln r ; \text{ This is in } 0 \cdot \infty \text{ form } \rightarrow 0 = \frac{1}{\infty} \text{ or } \infty = \frac{1}{0} \left[r^3 = \frac{1}{\frac{1}{r^3}} \right]$$

$$= 4 \cos^2 \theta \sin \theta \lim_{r \rightarrow 0^+} \frac{\ln r}{\frac{1}{r^3}} ; \text{ this is in } \frac{\infty}{\infty}, \text{ so apply L'Hopitals}$$

$$= 0$$

Example: 12 (A) Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} x^2 y \ln \sqrt{x^2 + y^2}$.

Example: 12 (B) Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} (2 + 3x) \ln(x^2 + y^2)^{\frac{5}{7}}$

Solution: $\lim_{(x,y) \rightarrow (0,0)} (2 + 3x) \ln(x^2 + y^2)^{\frac{5}{7}}$

$$= \lim_{(x,y) \rightarrow (0,0)} 2 \ln(x^2 + y^2)^{\frac{5}{7}} + \lim_{(x,y) \rightarrow (0,0)} 3x \ln(x^2 + y^2)^{\frac{5}{7}}$$

$$= \lim_{r \rightarrow 0^+} \frac{20}{7} \ln r + \lim_{r \rightarrow 0^+} \frac{30}{7} \cos \theta r \ln r = -\infty,$$

Limit does not exist. $-1 \leq \cos \theta \leq 1$

Note: $\lim_{r \rightarrow 0^+} \ln r = -\infty$.

Example: 13 Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} x^2 y \ln xy$

[Note: we need to apply $\ln(AB) = \ln A + \ln B$ after polar substitution.]

$$\begin{aligned} \text{Solution: } \lim_{(x,y) \rightarrow (0,0)} x^2 y \ln xy &= \lim_{r \rightarrow 0^+} \cos^2 \theta \sin \theta r^3 \ln(r^2 \sin \theta \cos \theta) \\ &= \lim_{r \rightarrow 0^+} \cos^2 \theta \sin \theta r^3 [\ln(r^2) + \ln(\sin \theta \cos \theta)] \\ &= \lim_{r \rightarrow 0^+} \cos^2 \theta \sin \theta r^3 \ln(r^2) + \lim_{r \rightarrow 0^+} \cos^2 \theta \sin \theta r^3 \ln(\sin \theta \cos \theta) \\ &= \cos^2 \theta \sin \theta \lim_{r \rightarrow 0^+} r^3 \ln r^2 + \cos^2 \theta \sin \theta \ln(\sin \theta \cos \theta) \lim_{r \rightarrow 0^+} r^3 \end{aligned}$$

If θ is the 2nd or in the 4th quadrant, then $\sin \theta \cos \theta < 0$. Hence, $\ln(\sin \theta \cos \theta)$ is not defined for all $0 \leq \theta \leq 2\pi$. Hence limit does not exist.

Note: Consider

$$\begin{aligned} &\lim_{r \rightarrow 0^+} \cos^2 \theta \sin \theta r^3 \ln(\sin \theta \cos \theta) \\ (A) \quad \lim_{r \rightarrow 0^+} \cos^2 \theta \sin \theta r^3 &= \cos^2 \theta \sin \theta \lim_{r \rightarrow 0^+} r^3 = (\cos^2 \theta \sin \theta)(0) = 0 \end{aligned}$$

$$-1 \leq \sin \theta \leq 1 \quad \& \quad -1 \leq \cos \theta \leq 1 \quad \text{implies} \quad -1 \leq \cos^2 \theta \sin \theta \leq 1 \quad \text{for } 0 \leq \theta \leq 2\pi.$$

$$(B) \quad \lim_{r \rightarrow 0^+} \ln(\sin \theta \cos \theta) = \lim_{r \rightarrow 0^+} [\ln(\sin \theta) + \ln(\cos \theta)]; \text{ where } 0 \leq \theta \leq 2\pi.$$

Example 14: Evaluate the limit $\lim_{(x,y) \rightarrow (0,0)} \frac{3yx}{x^2 + 2y^2}$

In polar coordinate:

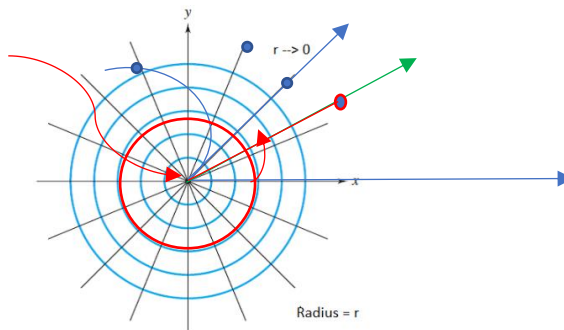
$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{3yx}{x^2 + 2y^2} &= \lim_{r \rightarrow 0^+} \frac{3(r \sin \theta)(r \cos \theta)}{(r \cos \theta)^2 + 2(r \sin \theta)^2} \quad ; \quad \text{where } 0 \leq \theta \leq 2\pi \\ &= \frac{3 \sin \theta \cos \theta}{\cos^2 \theta + 2 \sin^2 \theta} \lim_{r \rightarrow 0^+} \frac{r^2}{r^2} \\ &= \frac{3 \sin \theta \cos \theta}{\cos^2 \theta + 2 \sin^2 \theta} \lim_{r \rightarrow 0^+} 1 \end{aligned}$$

$$= \frac{3 \sin \theta \cos \theta}{\cos^2 \theta + 2 \sin^2 \theta} ; \text{ where } 0 \leq \theta \leq 2\pi$$

$$= \begin{cases} \frac{0}{1} & \text{for } \theta = 0 \\ \frac{3 \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}}}{\frac{1}{2} + 2 \left(\frac{1}{2} \right)} & \text{for } \theta = \frac{\pi}{4} \end{cases}$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} \frac{3yx}{x^2 + 2y^2} = \begin{cases} 0 & \text{for } \theta = 0 \text{ (along the curve } y = 0 \text{)} \\ 1 & \text{for } \theta = \frac{\pi}{4} \text{ (along the curve } y = x \text{)} \end{cases}$$

Hence, the limit does not exist.



$\frac{1}{0}(0) = 1???????? \rightarrow$ we don't allow this.

Definition: Continuous Function

A function $f(x, y)$ is said to be **continuous at** (a, b) if $f(a, b)$ is defined and if

$$\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b).$$

In addition, if f is continuous at every point in an open set D , then we say that f is **continuous on D** , and if f is continuous at every point in the xy -plane, then we say that f is **continuous everywhere**.

Recognizing Continuous Functions

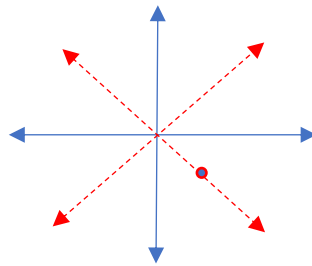
- A composition of continuous functions is continuous.
- A sum, difference, or product of continuous functions is continuous.
- A quotient of continuous functions is continuous, except where the denominator is zero.

Theorem:

- (a) A polynomial is continuous on its domain [domain is the whole space]
(b) A rational function is continuous on its domain [rational function is quotient of two polynomials,
Domain= The whole space, except those points where the denominator is zero]

Example: 13 Evaluate the limit $\lim_{(x,y) \rightarrow (1,-1)} \frac{yx}{x^2-y^2}$.

Domain: $D = \{(x,y): x^2 \neq y^2, i.e., x \neq \pm y\}$



Note that, if $f(x,y) = \frac{yx}{x^2-y^2}$, then $f(1,-1)$ is undefined. Hence, the function is not continuous at $(1,-1)$.

Definition:

Let f be a function of three variables, and assume that f is defined at all points within a ball centered at (x_0, y_0, z_0) , except possibly at (x_0, y_0, z_0) . We will write

$$\lim_{(x,y,z) \rightarrow (x_0,y_0,z_0)} f(x,y,z) = L$$

if given any number $\varepsilon > 0$, we can find a number $\delta > 0$ such that $f(x,y,z)$ satisfies

$$|f(x,y,z) - L| < \varepsilon$$

whenever the distance between (x,y,z) and (x_0,y_0,z_0) satisfies

$$0 < \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}$$

Note: If $x = r \cos \theta$, $y = r \sin \theta$, then $x^2 + y^2 = r^2 \rightarrow r = k$, $k > 0$ gives us a circle if θ has a full rotation, for example $0 \leq \theta \leq 2\pi$. If we get $\frac{1}{\sin \theta}$, then along the ray $\theta = \pi$, $\frac{1}{\sin \pi}$ is undefined. Hence, limit does not exist.

[13.2 is done]

Section 13.3: Partial Derivatives

$$\text{MAT-250: } z = f(x, y), \quad w = f(x, y, z)$$

Recall: $y = f(x)$:

Ordinary derivative or **derivative** is the derivative of a single variable function.

$$\text{Derivative of } f \text{ at } x: f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}; \quad h \neq 0.$$

$$\text{At } x = a: \frac{d}{dx}[f(x)]_{x=a} = f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}; \quad h \neq 0.$$

$$\frac{d}{dx}[2x^2 + 3] = 4x$$

$$\frac{d}{dx}[kx^2 + mx + p] = 2kx + m$$

$$\begin{aligned} \frac{d}{dx}[x^2y^2 + xy + 2] &= y^2 \frac{d}{dx}(x^2) + y \frac{d}{dx}(x) + 0; \quad y \text{ is a constant} \\ &= 2xy^2 + y \end{aligned}$$

$$\begin{aligned} \text{(A)} \quad \frac{d}{dx}[x^2y^2 + xy + 2]; \quad x \text{ is a constant } [x = 1] \\ &= 2x^2y + x \end{aligned}$$

$$\begin{aligned} \text{(B)} \quad \frac{d}{dx}[x^3y^2 + 3x^2y + 2xy^2]; \quad x \text{ is a constant} \\ &= 2x^3y + 3x^2 + 4xy \end{aligned}$$

$$\begin{aligned} \text{(C)} \quad \frac{d}{dx}[x^3y^2 + 3x^2y + 2xy^2]; \quad y \text{ is a constant} \\ &= 3x^2y^2 + 6xy + 2y^2 \end{aligned}$$

$$\text{If } y \text{ is a constant, then } \frac{d}{dx}[2x^2y^2 + 3y] = 4xy^2 + 0;$$

$$\text{If } x \text{ is a constant, then } \frac{d}{dy}[2x^2y^2 + 3y] = 4x^2y + 3$$

$$\frac{d}{dx} \rightarrow \text{Differential operator for ordinary derivative}$$

$$\frac{\partial}{\partial x} \rightarrow \text{Differential operator for partial derivative}$$

$$\text{For given } f(x, y) = 2xy + 1: \frac{\partial}{\partial x}[f(x, y)] = 2y \quad \text{and} \quad \frac{\partial}{\partial y}[f(x, y)] = 2x$$

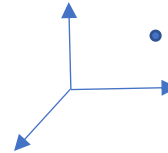
$$(c_1) \frac{d}{dx} [x^3 e^{3y} \cos(x^2 y^2)] ; y \text{ is constant}$$

$$= e^{3y} [3x^2 \cos(x^2 y^2) - 2x^4 y^2 \sin(x^2 y^2)]$$

$$\text{Formula: Product Rule} \rightarrow \frac{d}{dx} [f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$$

$$(c_2) \frac{d}{dy} [x^3 e^{3y} \cos(x^2 y^2)] ; x \text{ is constant}$$

$$= x^3 [3e^{3y} \cos(x^2 y^2) - e^{3y} 2x^2 y \sin(x^2 y^2)]$$



In 2D: $y = f(x) \rightarrow y$ is a constant, $y = c \leftarrow$ horizontal line in the xy -plane, perpendicular to y -axis and parallel to the x -axis.

In 3D: $z = f(x, y) \rightarrow y$ is a constant, $y = c \leftarrow$ this is a plane perpendicular to the y -axis in the xyz -space, parallel to the xz -plane.

[After Eid](#)

PARTIAL DERIVATIVES OF FUNCTIONS OF TWO VARIABLES $z = f(x, y)$

If $z = f(x, y)$, then one can inquire how the value of z changes if y is held fixed and x is allowed to vary, or if x is held fixed and y is allowed to vary. For example, the ideal gas law in physics states that under appropriate conditions the pressure exerted by a gas is a function of the volume of the gas and its temperature. Thus, a physicist studying gases might be interested in the rate of change of the pressure if the volume is held fixed and the temperature is allowed to vary, or if the temperature is held fixed and the volume is allowed to vary. We now define a derivative that describes such rates of change.

$$z = f(x, y) = 3x^2 y - x - y + 1$$

➔ The graph is in 3D. It's a surface.

If $y = y_0 = 2$, then $z = f(x, 2) = 6x^2 - x - 1 \leftarrow 2D$, a curve in the plane $y = 2$ parallel to xz - plane.

$\frac{dz}{dx} = \frac{\partial z}{\partial x} = \frac{\partial f(x, 2)}{\partial x} = 12x - 1$. This is the rate of change of z at any $(x, 2)$.

If $x = x_0 = 3$: $\frac{\partial z}{\partial x} \big|_{x=3} = \frac{\partial f(x, 2)}{\partial x} \big|_{x=3} = 12(3) - 1 = 35$.

The rate of change of f with respect to x at $(3, 2)$ is 35 and the rate of change of f with respect to x at $(-1, 2)$ is -13 .

Suppose that (x_0, y_0) is a point in the domain of a function $z = f(x, y)$. If we fix $y = y_0$, then $z = f(x, y_0)$ is a function of the variable x alone. The value of the derivative

$$\frac{d}{dx}[f(x, y_0)]$$

at $x = x_0$ then gives us a measure of the instantaneous rate of change of f with respect to x at the point (x_0, y_0) . Similarly, the value of the derivative

$$\frac{d}{dy}[f(x_0, y)]$$

at $y = y_0$ gives us a measure of the instantaneous rate of change of f with respect to y at the point (x_0, y_0) . These derivatives are so basic to the study of differential calculus of multivariable functions that they have their own name and notation.

$$[f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h} ; h \neq 0, \text{ Interval} = [x, x+h] \text{ or } [x+h, x], \text{ length} = |x+h-x| = |h|]$$

At $x = a$:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h)-f(a)}{h} ; h \neq 0, \text{ Interval} = [a, a+h] \text{ or } [a+h, a], \text{ length} = |a+h-a| = |h|]$$

$$\frac{\partial}{\partial x}[f(x, y_0)] = f_x(x, y_0) = \lim_{h \rightarrow 0} \frac{f(x+h, y_0)-f(x, y_0)}{h} ; h \neq 0. \quad h = (x+h) - x = \Delta x \rightarrow 0$$

Evaluate $f_x(x, y_0)$ at $x = x_0$:

$$f_x(x_0, y_0) = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} ; \Delta x \neq 0 \dots \dots \dots (1)$$

Similarly,

$$f_y(x_0, y_0) = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y} ; \Delta y \neq 0 \dots \dots \dots (2)$$

13 DEFINITION If $z = f(x, y)$ and (x_0, y_0) is a point in the domain of f , then the *partial derivative of f with respect to x* at (x_0, y_0) [also called the *partial derivative of z with respect to x* at (x_0, y_0)] is the derivative at x_0 of the function that results when $y = y_0$ is held fixed and x is allowed to vary. This partial derivative is denoted by $f_x(x_0, y_0)$ and is given by

$$f_x(x_0, y_0) = \left. \frac{d}{dx}[f(x, y_0)] \right|_{x=x_0} = \lim_{\Delta x \rightarrow 0} \frac{f(x_0 + \Delta x, y_0) - f(x_0, y_0)}{\Delta x} \quad (1)$$

Similarly, the *partial derivative of f with respect to y* at (x_0, y_0) [also called the *partial derivative of z with respect to y* at (x_0, y_0)] is the derivative at y_0 of the function that results when $x = x_0$ is held fixed and y is allowed to vary. This partial derivative is denoted by $f_y(x_0, y_0)$ and is given by

$$f_y(x_0, y_0) = \left. \frac{d}{dy}[f(x_0, y)] \right|_{y=y_0} = \lim_{\Delta y \rightarrow 0} \frac{f(x_0, y_0 + \Delta y) - f(x_0, y_0)}{\Delta y} \quad (2)$$

THE PARTIAL DERIVATIVE FUNCTIONS

Formulas (1) and (2) define the partial derivatives of a function at a specific point (x_0, y_0) . However, often it will be desirable to omit the subscripts and think of the partial derivatives as functions of the variables x and y . These functions are

$$f_x(x, y) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

$$f_y(x, y) = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

MAT-120: Example: 0

$$\frac{d}{dx}[5x^3 - 2x + 1] = 15x^2 - 2$$

$$\frac{d}{dx}[5p^2x^3 - 2q^5x + 11r] = 15p^2x^2 - 2q^5 + 0; \quad \text{where } p, q, r \text{ are constants}$$

$$\frac{d}{dx}[4m^3x^3 - 2nx + k] = 12m^3x^2 - 2n; \quad \text{where } m, n, k \text{ are constants.}$$

$$\frac{d}{dx}[5ax^3 - 2bx + c] = 15ax^2 - 2b$$

$$\frac{d}{dx}[5x^3 \sin y - 2xy^2 - 7y] = 15x^2 \sin y - 2y^2 - 0; \quad \text{where } y \text{ is a constant.}$$

$$\frac{d}{dx}(\sin(4x)) = \cos(4x) \frac{d}{dx}(4x) \frac{d}{dx}(x) = 4 \cos(4x)$$

Consider that y is a constant here in the following functions.

$$\frac{d}{dx}[5y^2x^3 - 2yx + 7y^3] = 15y^2x^2 - 2y; \quad \text{Note that } y \text{ is a constant here.}$$

1. If y is a constant, then

$$\begin{aligned} (a) \quad \frac{d}{dx}[y^2x^3e^{xy}] &= (y^2x^3) \frac{d}{dx}(e^{xy}) + e^{xy} \frac{d}{dx}(y^2x^3) \\ &= x^3y^3e^{xy} + 3x^2y^2e^{xy} = x^2y^2e^{xy}(xy + 3) \end{aligned}$$

$$(b) \quad \frac{d}{dx}[y^3x^3 \sin(x^2y^2)] = 2x^4y^5 \cos(x^2y^2) + 3x^2y^3 \sin(x^2y^2)$$

$$(c_1) \quad \frac{d}{dx}[x^3e^{3y} \cos(x^2y^2)]; \quad y \text{ is constant}$$

$$[\text{Formula: } \frac{d}{dx}(f(x)g(x)) = f(x)g'(x) + g(x)f'(x)]$$

$$= -2x^4y^2e^{3y} \sin(x^2y^2) + 3x^2e^{3y} \cos(x^2y^2)$$

$$(c_2) \quad \frac{d}{dy}[x^3e^{3y} \cos(x^2y^2)]; \quad x \text{ is constant}$$

$$= 3x^3e^{3y} \cos(x^2y^2) - 2x^5ye^{3y} \sin(x^2y^2)$$

$$(d) \quad \frac{d}{dx}[x^3e^{xy} \sin(2y)]$$

$$= \frac{d}{dx} [\sin(2y) (x^3 e^{xy})] = \sin(2y) \frac{d}{dx} [x^3 e^{xy}]$$

$$= \sin(2y) x^3 y e^{xy} + 3x^2 e^{xy} \sin(2y) = e^{xy} \sin(2y) [x^3 y + 3x^2]$$

2. If x is a constant, then

$$(a) \frac{d}{dy} [y^2 x^3 e^{xy}] = x^4 y^2 e^{xy} + 2x^3 y e^{xy} = x^3 y e^{xy} (xy + 2)$$

$$(b) \frac{d}{dy} [y^3 x^3 \sin(x^2 y^2)] = 2x^5 y^4 \cos(x^2 y^2) + 3x^3 y^2 \sin(x^2 y^2).$$

$$(c) \frac{d}{dy} [x^3 e^{3y} \cos(x^2 y^2)] = (x^3 e^{3y}) \frac{d}{dy} (\cos(x^2 y^2)) + \cos(x^2 y^2) \frac{d}{dy} (x^3 e^{3y})$$

$$= x^3 e^{3y} (-\sin(x^2 y^2)) (2x^2 y) + \cos(x^2 y^2) x^3 3e^{3y}$$

$$= -2x^5 y e^{3y} \sin(x^2 y^2) + 3x^3 e^{3y} \cos(x^2 y^2)$$

$$(d) \frac{d}{dy} [x^3 e^{xy} \sin(2y)] = x^3 [x \sin(2y) e^{xy} + 2e^{xy} \cos(2y)] = x^3 e^{xy} [2 \cos(2y) + x \sin(2y)]$$

Note that, $\frac{d}{dx} [c f(x)] = c \frac{d}{dx} [f(x)]$

Implicit: Function $f = \{(x, y)\}$, $f = \{(x, y, z)\}$, $f = \{(x, y, z, w)\}$

If $y = f(x)$, then find $\frac{dy}{dx}$ from $5y^2 x^3 - 2yx = -7y^3 \dots \rightarrow y = f(x)$ *implicit function*.

If $y = f(x)$, then $\frac{d}{dx} (y^3) = 3y^2 \frac{dy}{dx}$, Note: $\frac{dx}{dx} = 1$, but $\frac{dy}{dx} \neq 1$.

Notations: If $y = f(x)$, that is, x is the only input of the function, then the notations are:

$$\frac{dy}{dx}, \quad \frac{df}{dx}, \quad f'(x), \quad y', \quad \frac{d^2 f}{dx^2} = f^{(2)}(x) = f'', \quad \frac{d^4 f}{dx^4} = f^{IV}$$

THE PARTIAL DERIVATIVE NOTATIONS FOR THE FUNCTION $z = f(x, y)$

$$\frac{\partial f}{\partial x} = \frac{\partial z}{\partial x} = f_x = z_x \quad \text{and} \quad \frac{\partial f}{\partial y} = \frac{\partial z}{\partial y} = f_y = z_y$$

HIGHER ORDER PARTIAL DERIVATIVES NOTATIONS FOR THE FUNCTION $z = f(x, y)$

Higher Order Partial Derivatives with respect to x : $\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2} = f_{xx}$, $\frac{\partial^4 f}{\partial x^4} = \frac{\partial^4 z}{\partial x^4} = f_{xxxx}$,

Higher Order Partial Derivatives with respect to y : $\frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2} = f_{yy}$, $\frac{\partial^4 f}{\partial y^4} = \frac{\partial^4 z}{\partial y^4} = f_{yyyy}$,

HIGHER ORDER MIXED PARTIAL DERIVATIVES NOTATIONS FOR THE FUNCTION $z = f(x, y)$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (f_x) = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial y} (z_x) = f_{xy} = \frac{\partial}{\partial y} (f_x)$$

$$\frac{\partial^4 f}{\partial x \partial y^2 \partial x} = \frac{\partial^4 z}{\partial x \partial y^2 \partial x} = f_{xyyx}$$

$$w = f(x, y, z) : \frac{\partial^4 f}{\partial z \partial y^2 \partial x} = \frac{\partial^4 w}{\partial z \partial y^2 \partial x} = f_{xyyz}$$

Example: 1 (a) Find derivative functions $f_x(x, y)$ and $f_y(x, y)$ for $f(x, y) = 2x^3y^2 + 2y + 4x$, and use those partial derivatives to compute $f_x(1, 3)$ and $f_y(1, 3)$.

Given $f(x, y) = 2x^3y^2 + 2y + 4x$

$f_x(x, y) = 6x^2y^2 + 4$ and $f_y(x, y) = 4x^3y + 2$ and then we get $f_x(1, 3) = 58$ and $f_y(1, 3) = 14$.

Formula: $\frac{d}{dx}[e^{f(x)}] = e^{f(x)} f'(x)$

Example: 1 (b) Find derivative functions $f_x(x, y)$ and $f_y(x, y)$ for $f(x, y) = 2x^3e^{x^2}y^2 + 2y + 4x$, and use those partial derivatives to compute $f_x(1, 0)$ and $f_y(1, 0)$.

Solution:

Given $f(x, y) = 2x^3e^{x^2}y^2 + 2y + 4x$

$$\begin{aligned} f_x(x, y) &= \frac{\delta}{\delta x} (2x^3e^{x^2}y^2 + 2y + 4x) \\ &= 2y^2 \frac{\delta}{\delta x} (x^3e^{x^2}) + 0 + 4 = 2y^2 \left[x^3 \frac{\delta}{\delta x} (e^{x^2}) + e^{x^2} \frac{\delta}{\delta x} (x^3) \right] \\ &= 2y^2 [x^3 e^{x^2} 2x + e^{x^2} 3x^2] + 4 \\ &= 2y^2 [2x^4 e^{x^2} + 3x^2 e^{x^2}] + 4 \end{aligned}$$

$$f_y(x, y) = 4x^3e^{x^2}y + 2$$

$$f_x(1, 0) = 4$$

$$f_y(1, 0) = 2$$

Example: 2 (a) Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $z = x^4 \sin(xy^3)$.

$$\frac{\partial z}{\partial x} = 4x^3 \sin(xy^3) + x^4 y^3 \cos(xy^3)$$

$$\frac{\partial z}{\partial y} = 3x^5 y^2 \cos(xy^3)$$

Example: 2 (b) Find $\frac{\partial^2 z}{\partial x^2}$, $\frac{\partial^2 z}{\partial x \partial y}$ and $\frac{\partial^2 z}{\partial y^2}$ if $z = x^4 \sin(xy^3)$. **Homework**

Solution:

$$\text{Here } \frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (x^4 \sin(xy^3))$$

$$= x^4 \frac{\partial}{\partial x} (\sin(xy^3)) + \sin(xy^3) \frac{\partial}{\partial x} (x^4)$$

$$= x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3)$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} (x^4 y^3 \cos(xy^3) + 4x^3 \sin(xy^3))$$

$$= -x^4 y^6 \sin(xy^3) + 4x^3 y^3 \cos(xy^3) + 4x^3 y^3 \cos(xy^3) + 12x^2 \sin(xy^3)$$

$$= 12x^2 \sin(xy^3) + 8x^3 y^3 \cos(xy^3) - x^4 y^6 \sin(xy^3).$$

Complete at home!

Example: 3 (a) Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ if $z = e^{4x} \tan(x^2 y^3)$.

$$\frac{\partial z}{\partial x} = e^{4x} \frac{\partial}{\partial x} (\tan(x^2 y^3)) + \tan(x^2 y^3) \frac{\partial}{\partial x} (e^{4x})$$

$$= e^{4x} \sec^2(x^2 y^3) 2xy^3 + \tan(x^2 y^3) 4e^{4x}$$

$$= 2e^{4x} [xy^3 \sec^2(x^2 y^3) + 2 \tan(x^2 y^3)]$$

$$\frac{\partial z}{\partial y} = 3x^2 y^2 e^{4x} \sec^2(x^2 y^3)$$

(b) Find $f_{xx}(x, y)$, $f_{yy}(x, y)$ and $f_{xy}(x, y)$ for $f(x, y) = e^{4x} \tan(x^2 y^3)$.

$$f_y(x, y) = e^{4x} \frac{\partial}{\partial y} (\tan(x^2 y^3)) =$$

$$\text{Now } f_{yy}(x, y) = \frac{\partial}{\partial y} (f_y(x, y)) = \frac{\partial}{\partial y} (3x^2 y^2 e^{4x} \sec^2(x^2 y^3))$$

$$= 3x^2 e^{4x} \frac{\partial}{\partial y} (y^2 \sec^2(x^2 y^3)) = 3x^2 e^{4x} \left[y^2 \frac{\partial}{\partial y} (\sec^2(x^2 y^3)) + \sec^2(x^2 y^3) \frac{\partial}{\partial y} (y^2) \right]$$

$$\begin{aligned}
&= 3x^2 e^{4x} [y^2 \cdot 2 \sec(x^2 y^3) \cdot \sec(x^2 y^3) \tan(x^2 y^3) \cdot 3x^2 y^2 + 2y \sec^2(x^2 y^3)] \\
&= 3x^2 e^{4x} [6x^2 y^4 \sec^2(x^2 y^3) \tan(x^2 y^3) + 2y \sec^2(x^2 y^3)] \\
&= 6x^2 y e^{4x} \sec^2(x^2 y^3) [3x^2 y^3 \tan(x^2 y^3) + 1].
\end{aligned}$$

- Find $f_{xx}(x, y)$ for $f(x, y) = e^{4x} \tan(x^2 y^3)$.

Here

$$\begin{aligned}
f_x(x, y) &= \frac{\partial}{\partial x} (e^{4x} \tan(x^2 y^3)) = e^{4x} \frac{\partial}{\partial x} (\tan(x^2 y^3)) + \tan(x^2 y^3) \frac{\partial}{\partial x} (e^{4x}) \\
&= 4e^{4x} \tan(x^2 y^3) + 2xy^3 e^{4x} \sec^2(x^2 y^3) \\
f_{xx}(x, y) &= \frac{\partial}{\partial x} (4e^{4x} \tan(x^2 y^3) + 2xy^3 e^{4x} \sec^2(x^2 y^3)) \text{ complete!}
\end{aligned}$$

Example: 4 (a) Find $f_{xx}(x, y)$, $f_{yy}(x, y)$ and $f_{xy}(x, y)$ for $f(x, y) = 2x^3 y^2 + 2y + 4x$, and use those partial derivatives to compute $f_{xx}(1, 3)$, $f_{yy}(1, 3)$ and $f_{xy}(1, 3)$

(b) Find $f_{xx}(x, y)$, $f_{yy}(x, y)$ and $f_{xy}(x, y)$ for $f(x, y) = 5x^3 e^y + 2y + 4x$, and use those partial derivatives to compute $f_{xx}(1, 0)$, $f_{yy}(1, 0)$ and $f_{xy}(1, 0)$

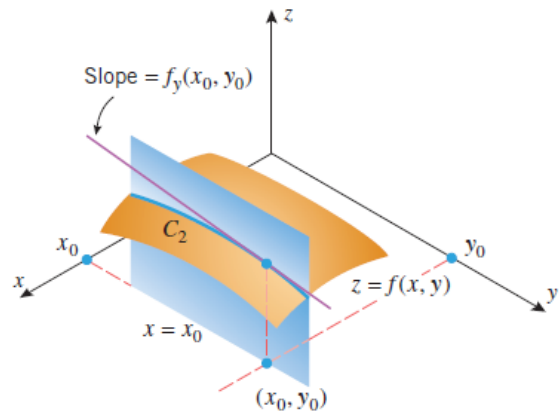
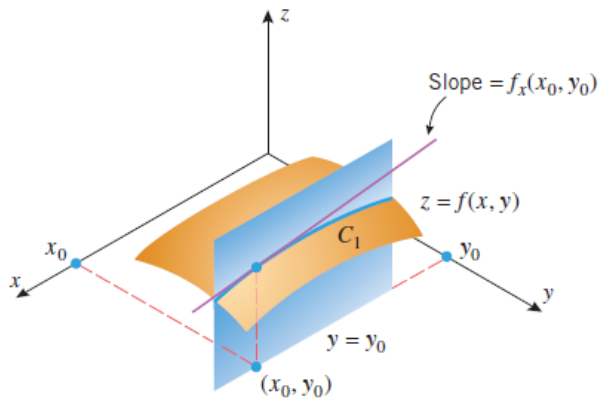
(c) Find $f_{xx}(x, y)$, $f_{yy}(x, y)$ and $f_{xy}(x, y)$ for $f(x, y) = 5x^3 \cos y - 2xy + 4x^2 y^2$, and use those partial derivatives to compute $f_{xx}(1, 0)$, $f_{yy}(1, 0)$ and $f_{xy}(1, 0)$

(d) Find $f_{xx}(x, y)$, $f_{yy}(x, y)$ and $f_{xy}(x, y)$ for $f(x, y) = 5x^3 \cos y + 2x^2 y + 4xy$, and use those partial derivatives to compute $f_{xx}(1, 0)$, $f_{yy}(1, 0)$ and $f_{xy}(1, 0)$.

THEOREM Let f be a function of two variables. If f_{xy} and f_{yx} are continuous on some open disk, then $f_{xy} = f_{yx}$ on that disk.

PARTIAL DERIVATIVES VIEWED AS RATES OF CHANGE AND SLOPES: $z = f(x, y)$

$f_x(x, y_0)$ can be interpreted as the rate of change of z with respect to x along the curve C_1 , and $f_y(x_0, y)$ can be interpreted as the rate of change of z with respect to y along the curve C_2 . In particular, $f_x(x_0, y_0)$ is the rate of change of z with respect to x along the curve C_1 at the point (x_0, y_0) , and $f_y(x_0, y_0)$ is the rate of change of z with respect to y along the curve C_2 at the point (x_0, y_0) .



$z = f(x, y)$ & Think $y = y_0$ constant $z = f(x, y_0) \rightarrow$ 1 Dimensional Curve C_1

$C_1 \rightarrow$ intersection of the surface $z = f(x, y)$ and the plane $y = y_0$.

Review 3.1: If $y = f(x)$, then find the implicit derivative $\frac{dy}{dx}$ from $5y^2x^3 - 2yx = -7y^3$

Given, $5y^2x^3 - 2yx = -7y^3$

Taking derivative of both sides with respect to x :

$$\frac{d}{dx} [5y^2x^3 - 2yx] = \frac{d}{dx} (-7y^3); \text{ Not partial derivative}$$

$$\Rightarrow 15x^2y^2 + 10x^3y \frac{dy}{dx} - 2\left(y + x \frac{dy}{dx}\right) = -21y^2 \frac{dy}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{2y - 15x^2y^2}{10x^3y + 21y^2 - 2x}$$

$$f = \{(x, y, z)\} \rightarrow z = f(x, y)$$

Example: 2 : Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ from $5x^3y^2z^2 - 2xyz = -7z$

To find $\frac{\partial z}{\partial x}$:

$$\frac{\partial}{\partial x} [5x^3y^2z^2 - 2xyz] = \frac{\partial}{\partial x} [-7z]$$

$$\frac{\partial}{\partial x} [(5y^2x^3)z^2 - (2yx)z] = \frac{\partial}{\partial x} [-7z]$$

$$\Rightarrow 5y^2x^3 \frac{\partial}{\partial x} (z^2) + z^2 \frac{\partial}{\partial x} (5y^2x^3) - \left[2yx \frac{\partial}{\partial x} (z) + z \frac{\partial}{\partial x} (2yx)\right] = -7 \frac{\partial z}{\partial x}$$

$$\begin{aligned}
&\Rightarrow 5y^2x^3 2z \frac{\partial z}{\partial x} + z^2 15x^2y^2 - \left[2yx \frac{\partial z}{\partial x} + z 2y \right] = -7 \frac{\partial z}{\partial x} \\
&\Rightarrow 10x^3y^2z \frac{\partial z}{\partial x} + 15x^2y^2z^2 - 2xy \frac{\partial z}{\partial x} - 2yz = -7 \frac{\partial z}{\partial x} \\
&\Rightarrow 10x^3y^2z \frac{\partial z}{\partial x} - 2xy \frac{\partial z}{\partial x} + 7 \frac{\partial z}{\partial x} = 2yz - 15x^2y^2z^2 \\
&\Rightarrow (10x^3y^2z - 2xy + 7) \frac{\partial z}{\partial x} = 2yz - 15x^2y^2z^2
\end{aligned}$$

$$\therefore \frac{\partial z}{\partial x} = \frac{2yz - 15x^2y^2z^2}{(10x^3y^2z - 2xy + 7)}$$

and $\frac{\partial z}{\partial y} = ????$

Please complete !

Section: 13.3 Please practice following exercises

3, 5, 7, 11, 13, 27, 29, 30, 32, 33, 34,35, 36, 43, 44,45, 46, 48,49, 55, 58, 61, 67, 69,71, 74, 76,86, 89, 90, 91, 92,101, 102, 104, 106

Section 13.5: Chain Rules

Recall: If y is a differentiable function of x ($y = f(x)$) and x is a differentiable function of t ($x = x(t)$), then the chain rule for function of one variable states that, under composition $y = f(x(t))$, y becomes a differentiable function of t with

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} \quad \text{Or} \quad \frac{df}{dt} = \frac{df}{dx} \frac{dx}{dt} = f'(x(t)) \cdot x'(t).$$

For the functions $z = f(x, y)$ and $w = f(x, y, z)$ the Chain Rule for Derivatives:

CHAIN RULES FOR (ORDINARY) DERIVATIVES

13.5.1 THEOREM (Chain Rules for Derivatives) If $x = x(t)$ and $y = y(t)$ are differentiable at t , and if $z = f(x, y)$ is differentiable at the point $(x, y) = (x(t), y(t))$, then $z = f(x(t), y(t))$ is differentiable at t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \quad (5)$$

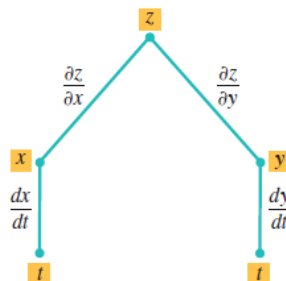
where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y) .

If each of the functions $x = x(t)$, $y = y(t)$, and $z = z(t)$ is differentiable at t , and if $w = f(x, y, z)$ is differentiable at the point $(x, y, z) = (x(t), y(t), z(t))$, then the function $w = f(x(t), y(t), z(t))$ is differentiable at t and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt} \quad (6)$$

where the ordinary derivatives are evaluated at t and the partial derivatives are evaluated at (x, y, z) .

Tree for $z = f(x, y)$, $x = x(t)$, $y = y(t)$: for $z = f(x(t), y(t))$



$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$

Example: 1 (a) Consider $z = f(x, y) = 3x^2y^4$; $x = x(t) = 2t^2$, $y = y(t) = t^3$. Find $\frac{dz}{dt}$.

There are 2-different methods.

Method-1: By the Chain Rule, we know that $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \dots \dots \dots$ (1)

Here, $z = 3x^2y^4$, $x = 2t^2$, $y = t^3$.

So, $\frac{\partial z}{\partial x} = 6xy^4$, $\frac{\partial z}{\partial y} = 12x^2y^3$, $\frac{dx}{dt} = 4t$, $\frac{dy}{dt} = 3t^2$

From (1): $\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} = 6xy^4 \cdot 4t + 12x^2y^3 \cdot 3t^2$

$\Rightarrow \frac{dz}{dt} = 24t xy^4 + 36t^2 x^2y^3$
 $= 24t (2t^2)(t^3)^4 + 36t^2 (2t^2)^2 (t^3)^3$; substitute: $x = 2t^2$, $y = t^3$.
 $= 48t^{15} + 144t^{15} = 192t^{15}$

$$\frac{dz}{dt} = 192t^{15}$$

Method-2: Here $z = 3x^2y^4$; $x = 2t^2$, $y = t^3$.

Substitute: $x = 2t^2$, $y = t^3$ in z . So, $z = 3(2t^2)^2(t^3)^4 \rightarrow z = 12t^{16}$

$$\frac{dz}{dt} = 12 \times 16 t^{15} = 192 t^{15}.$$

Example: 1 (b) Consider $z = \ln(x^2y)$; $x = \sin t$, $y = \tan t$. Find $\frac{dz}{dt}$. $\rightarrow$ Homework

(c) Consider $z = \ln(x^2y + 3x^2)$; $x = \sin t$, $y = \tan t$. Find $\frac{dz}{dt}$. $\rightarrow$ Homework

(d) Consider $z = \sin(x^2y)$; $x = \sin t$, $y = \tan t$. Find $\frac{dz}{dt}$. $\rightarrow$ Homework

CHAIN RULES FOR PARTIAL DERIVATIVES

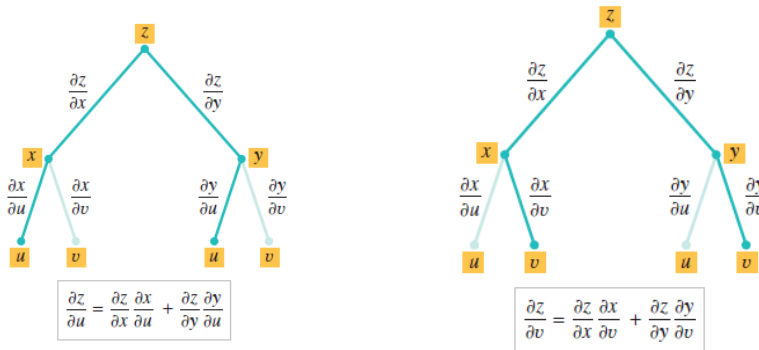
13.5.2 THEOREM (Chain Rules for Partial Derivatives) If $x = x(u, v)$ and $y = y(u, v)$ have first-order partial derivatives at the point (u, v) , and if $z = f(x, y)$ is differentiable at the point $(x, y) = (x(u, v), y(u, v))$, then $z = f(x(u, v), y(u, v))$ has first-order partial derivatives at the point (u, v) given by

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \quad \text{and} \quad \frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v} \quad (9-8)$$

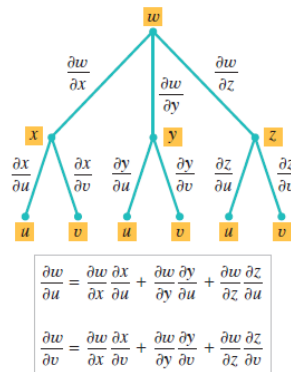
If each function $x = x(u, v)$, $y = y(u, v)$, and $z = z(u, v)$ has first-order partial derivatives at the point (u, v) , and if the function $w = f(x, y, z)$ is differentiable at the point $(x, y, z) = (x(u, v), y(u, v), z(u, v))$, then $w = f(x(u, v), y(u, v), z(u, v))$ has first-order partial derivatives at the point (u, v) given by

$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial u} \quad \text{and} \quad \frac{\partial w}{\partial v} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial v} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial v} \quad (9-10)$$

Trees for $z = f(x, y)$:



Tree for $w = f(x, y, z)$:



→ **THINK: Check This:** $\frac{\partial}{\partial v} \left(\frac{\partial z}{\partial u} \right) = \frac{\partial}{\partial v} \left[\frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \right] = \frac{\partial}{\partial v} \left[\frac{\partial z}{\partial x} \frac{\partial x}{\partial u} \right] + \frac{\partial}{\partial v} \left[\frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \right] = ?$ Use product Rule to complete.

Example: 2 (a) Consider $z = 3x^2y^4$; $x = 2u + v^2$, $y = \frac{u}{v}$. Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.

We know that $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}$ and $\left[\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v} \right] \leftarrow \text{Homework}$

Solution:

We know that $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} \dots \dots (1)$

Given, $z = 3x^2y^4$; $x = 2u + v^2$, $y = \frac{u}{v} = \frac{1}{v}u = u \frac{1}{v} = u v^{-1}$.

So, $\frac{\partial z}{\partial x} = 6xy^4$; $\frac{\partial x}{\partial u} = 2$; $\frac{\partial z}{\partial y} = 12x^2y^3$; $\frac{\partial y}{\partial u} = \frac{1}{v}$

$$\begin{aligned} \text{From (1): } \frac{\partial z}{\partial u} &= (6xy^4)(2) + (12x^2y^3) \left(\frac{1}{v} \right) = 12xy^4 + \frac{12}{v} x^2y^3 \\ &= 12(2u + v^2) \left(\frac{u}{v} \right)^4 + \frac{12}{v} (2u + v^2)^2 \left(\frac{u}{v} \right)^3; [\text{substitute } x = 2u + v^2, y = \frac{u}{v}] \\ &= \frac{12u^4}{v^4} (2u + v^2) + \frac{12u^3}{v^4} (2u + v^2)^2 \\ &= \frac{12u^3}{v^4} (2u + v^2) [u + 2u + v^2] \end{aligned}$$

$$\frac{\partial z}{\partial u} = \frac{12u^3}{v^4} (2u + v^2)(3u + v^2).$$

Example: 2(b) Consider $z = 3x^2y^4$; $x = 2u + v^2 + w$, $y = \frac{u}{v} - w^2$. Find $\frac{\partial z}{\partial w}$

$$\frac{\partial z}{\partial u}, \quad \frac{\partial z}{\partial v} \quad \text{and} \quad \frac{\partial z}{\partial w} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial w} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial w}$$

Example: 2 (c) Consider $z = 3x^2y^4w^3$; $x = 2u + v^2$, $y = \frac{u}{v}$, $w = uv$.

$$\text{Here, } \frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} + \frac{\partial z}{\partial w} \frac{\partial w}{\partial u}$$

Example: 3 Consider $w = x^2 + y^2 - z^2$; where $x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta$

and $z = \rho \cos \varphi$. Find $\frac{\partial w}{\partial \rho}$, $\frac{\partial w}{\partial \varphi}$ and $\frac{\partial w}{\partial \theta}$.

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \theta} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial \theta}$$

We need: $\frac{\partial w}{\partial x} = 2x$, $\frac{\partial w}{\partial y} = 2y$, $\frac{\partial w}{\partial z} = -2z$

$$\frac{\partial x}{\partial \theta} = -\rho \sin \varphi \sin \theta, \quad \frac{\partial y}{\partial \theta} = \rho \sin \varphi \cos \theta \quad \text{and} \quad \frac{\partial z}{\partial \theta} = 0.$$

$$\frac{\partial w}{\partial \theta} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \theta} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial \theta} \dots \dots (1)$$

Substitute in Equation (1):

$$\begin{aligned}\frac{\partial w}{\partial \theta} &= 2x(-\rho \sin \varphi \sin \theta) + 2y(\rho \sin \varphi \cos \theta) - 2z(0) \\ &= 2(\rho \sin \varphi \cos \theta)(-\rho \sin \varphi \sin \theta) + 2(\rho \sin \varphi \sin \theta)(\rho \sin \varphi \cos \theta) \\ &= 0\end{aligned}$$

Homework: Find $\frac{\partial w}{\partial \rho}$ and $\frac{\partial w}{\partial \varphi}$

Given $w = x^2 + y^2 - z^2$; where $x = \rho \sin \varphi \cos \theta$, $y = \rho \sin \varphi \sin \theta$, and $z = \rho \cos \varphi$.

Find $\frac{\partial w}{\partial \varphi}$

We know that $\frac{\partial w}{\partial \varphi} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial \varphi} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial \varphi} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial \varphi} \dots \dots \dots (1)$

$$\frac{\partial w}{\partial x} = 2x, \quad \frac{\partial w}{\partial y} = 2y, \quad \frac{\partial w}{\partial z} = -2z$$

$$\frac{\partial x}{\partial \varphi} = \rho \cos \varphi \cos \theta, \quad \frac{\partial y}{\partial \varphi} = \rho \cos \varphi \sin \theta, \quad \frac{\partial z}{\partial \varphi} = -\rho \sin \varphi.$$

$$\frac{\partial w}{\partial \varphi} = 2(\rho \sin \varphi \cos \theta)(\rho \cos \varphi \cos \theta) + 2(\rho \sin \varphi \sin \theta)(\rho \cos \varphi \sin \theta) - 2(\rho \cos \varphi)(-\rho \sin \varphi)$$

Please simplify !

Example:4

(a) Consider $z = x^2 + y^2$; where $x = r \cos \theta + 1$, $y = r \sin \theta$. Find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$. Please submit

(b) Consider $z = \sqrt{x^2 + y^2}$; where $x = r \cos \theta$, $y = r \sin \theta$. Find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$

(c) Consider $z = (x^2 + y^2)^{\frac{1}{3}}$; where $x = r \cos \theta$, $y = r \sin \theta$. Find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$

CHAIN RULES FOR IMPLICIT DIFFERENTIATION

Common practice: Implicit functions are given as equations.

Equation of two variables $(x, y) \leftarrow$ equation can define y as an implicit function of x .

Equation of three variables $(x, y, z) \leftarrow$ equation can define z as an implicit function of x and y .

Equation of three variables $(x, y, z, w) \leftarrow$ equation can define w as an implicit function of x, y and z .

Note that, we can choose any one of those variables as the output, and other(s) as input(s).

13.5.3 THEOREM *If the equation $f(x, y) = c$ defines y implicitly as a differentiable function of x , and if $\partial f / \partial y \neq 0$, then*

$$\frac{dy}{dx} = -\frac{\partial f / \partial x}{\partial f / \partial y} \quad (14)$$

Example: 5

(a) Find the implicit derivative $\frac{dy}{dx}$ from $5y^2x^3 - 2yx = -7y^3$.

(b) Find the implicit derivative $\frac{dy}{dx}$ from $5y^2x^3 - 2yx = -7y^3$ at the origin $(0,0)$.

(a) Solution: Given $5y^2x^3 - 2yx = -7y^3$, that is, $5y^2x^3 - 2yx + 7y^3 = 0 \dots (1)$.

Set $f(x, y) = 5y^2x^3 - 2yx + 7y^3$

From equation (1), we get $f(x, y) = 5y^2x^3 - 2yx + 7y^3 = 0$, that is, $f(x, y) = 0$.

Hence, $f(x, y) = 0$ defines y as an implicit differentiable function of x .

Here, $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(5y^2x^3 - 2yx + 7y^3)$.

$$= 5x^3 \frac{\partial}{\partial y}(y^2) - 2x \frac{\partial}{\partial y}(y) + 7 \frac{\partial}{\partial y}(y^3)$$

$$= 5x^3(2y) - 2x(1) + 7(3y^2)$$

$$\frac{\partial f}{\partial y} = 10x^3y - 2x + 21y^2$$

Note that $\frac{\partial f}{\partial y} = 10x^3y - 2x + 21y^2 \neq 0$ at least at $(1,1)$. Also, $\frac{\partial f}{\partial x} = 15y^2x^2 - 2y$

Then we get $\frac{dy}{dx} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} = -\frac{15y^2x^2-2y}{10yx^3-2x+21y^2}$.

13.5.4 THEOREM *If the equation $f(x, y, z) = c$ defines z implicitly as a differentiable function of x and y , and if $\partial f / \partial z \neq 0$, then*

$$\frac{\partial z}{\partial x} = -\frac{\partial f / \partial x}{\partial f / \partial z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{\partial f / \partial y}{\partial f / \partial z}$$

Example: 5

(b) Find the implicit derivatives $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ from $5x^3y^2z^2 - 2xyz = -7z + 1$.

Solution: Given, $5x^3y^2z^2 - 2xyz = -7z + 1$.

Hence, $5x^3y^2z^2 - 2xyz + 7z = 1$

Set $w = f(x, y, z) = 5x^3y^2z^2 - 2xyz + 7z$.

Then we get, $w = f(x, y, z) = 5x^3y^2z^2 - 2xyz + 7z = 1$, that is, $f(x, y, z) = 1$, constant. The equation $5x^3y^2z^2 - 2xyz + 7z = 1$ defines z as a function of x and y .

Also, for $f(x, y, z) = 5x^3y^2z^2 - 2xyz + 7z$,

$$\frac{\partial f}{\partial z} = 10x^3y^2z - 2xy + 7 \neq 0 \text{ at least at } (0,0,0).$$

Hence implicit derivatives of z :

$$\frac{\partial z}{\partial x} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial z}} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial z}}$$

$$\frac{\partial f}{\partial x} = 15x^2y^2z^2 - 2yz, \quad \frac{\partial f}{\partial y} = 10x^3yz^2 - 2xz$$

Then we get $\frac{\partial z}{\partial x} = -\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial z}} = -\frac{15x^2y^2z^2-2yz}{10x^3y^2z-2xy+7}$ and $\frac{\partial z}{\partial y} = -\frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial z}} = -\frac{10x^3yz^2-2xz}{10x^3y^2z-2xy+7}$

Note: Finding Asymptotes

$$f(x) = \frac{x^2 + 1}{(x - 2)(x - 3)} = \frac{x^2 + 1}{x^2 - 5x + 6}$$

$$\text{Set } y = \frac{x^2 + 1}{x^2 - 5x + 6}$$

$$\text{Horizontal Asymptote: } y = \lim_{x \rightarrow \pm\infty} \frac{x^2 + 1}{x^2 - 5x + 6} = \lim_{x \rightarrow \pm\infty} \frac{1 + \frac{1}{x^2}}{1 - \frac{5}{x} + \frac{6}{x^2}}, \text{ that is, } y = 1$$

$$\text{Vertical Asymptote: } x = 2, x = 3$$

Question: Evaluate

$$\int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} t \sin(k\omega t) dt; \quad k - \text{integer}, \quad \omega \neq 0.$$

Solution: If $k = 0$, then we get

$$\int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} t \sin(0) dt = \int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} 0 dt = 0.$$

Assume that $k \neq 0$. We start with the indefinite integral

$$\int t \sin(k\omega t) dt;$$

$$\text{Set } u = t, \quad dv = \sin(k\omega t) dt. \text{ Then we have } du = dt \quad \text{and} \quad v = \int \sin(k\omega t) dt = -\frac{\cos(k\omega t)}{k\omega}$$

$$= \int u dv$$

$$= uv - \int v du$$

$$= t \left(-\frac{\cos(k\omega t)}{k\omega} \right) - \int -\frac{\cos(k\omega t)}{k\omega} dt$$

$$= -\frac{t \cos(k\omega t)}{k\omega} + \frac{1}{k\omega} \int \cos(k\omega t) dt$$

$$= -\frac{t \cos(k\omega t)}{k\omega} + \frac{1}{k\omega} \left[\frac{\sin(k\omega t)}{k\omega} \right] + C$$

$$\int t \sin(k\omega t) dt = -\frac{t \cos(k\omega t)}{k\omega} + \frac{1}{k^2\omega^2} \sin(k\omega t) + C$$

Now for the definite integral,

$$\int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} t \sin(k\omega t) dt = \left[-\frac{t \cos(k\omega t)}{k\omega} + \frac{1}{k^2\omega^2} \sin(k\omega t) + C \right]_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}}$$

$$= -\frac{\frac{\pi}{\omega} \cos\left(k\omega \frac{\pi}{\omega}\right)}{k\omega} + \frac{1}{k^2\omega^2} \sin\left(k\omega \frac{\pi}{\omega}\right) + C - \left[-\frac{-\frac{\pi}{\omega} \cos\left(-k\omega \frac{\pi}{\omega}\right)}{k\omega} + \frac{1}{k^2\omega^2} \sin\left(-k\omega \frac{\pi}{\omega}\right) + C \right]$$

$$= -\frac{\frac{\pi}{\omega} \cos(k\pi)}{k\omega} + \frac{1}{k^2\omega^2} \sin(k\pi) - \frac{\frac{\pi}{\omega} \cos(k\pi)}{k\omega} - \frac{1}{k^2\omega^2} \sin(k\pi) - C$$

$$= -\frac{2 \frac{\pi}{\omega} \cos(k\pi)}{k\omega}; \quad [\cos(2k\pi) = 1 \text{ and } \cos(2k+1)\pi = -1 \text{ for any integer } k]$$

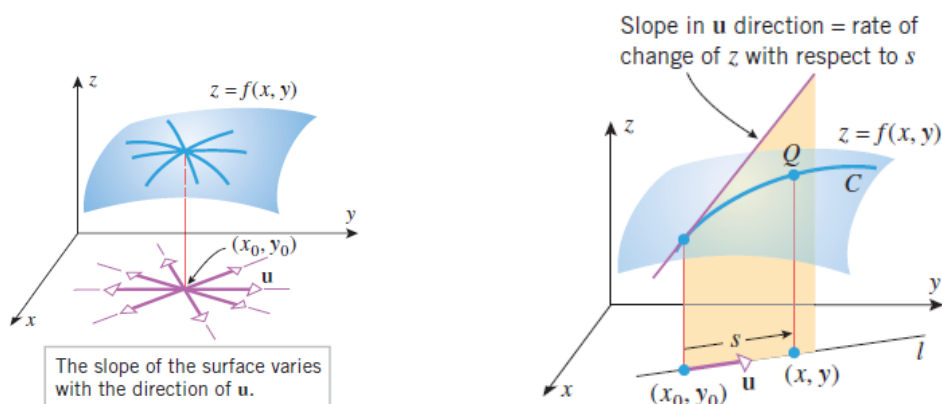
$$= -\frac{2 \pi}{k\omega^2} \cos(k\pi) ; \quad \cos(k\pi) = 1 \text{ if } k \text{ is even and } \cos(k\pi) = -1 \text{ if } k \text{ is odd.}$$

$$= -\frac{2 \pi}{k\omega^2} (-1)^k ; \quad \text{for any integer } k.$$

$$\int_{-\frac{\pi}{\omega}}^{\frac{\pi}{\omega}} t \sin(k\omega t) dt = \frac{2 \pi}{k\omega^2} (-1)^{k+1} ; \quad \text{for any integer } k. \quad [\text{since } a^k (a) = a^{k+1}]$$

Section 13.6 Directional Derivatives

Suppose that we wish to compute the instantaneous rate of change of a function $f(x, y)$ with respect to distance from a point (x_0, y_0) in some direction. Since there are infinitely many different directions from (x_0, y_0) in which we could move, we need a convenient method for describing a specific direction starting at (x_0, y_0) .



One way to do this is to use a unit vector $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ that has its initial point at (x_0, y_0) and points in the desired direction of $\mathbf{u}$. This vector determines a line l in the xy -plane that can be expressed parametrically as

$$x = x_0 + su_1, \quad y = y_0 + su_2$$

Since $\mathbf{u}$ is a unit vector, s is the arc length parameter that has its reference point at (x_0, y_0) and has positive values in the direction of $\mathbf{u}$. For $s = 0$, the point (x, y) is at the reference point (x_0, y_0) , and as s increases, the point (x, y) moves along l in the direction of $\mathbf{u}$. On the line l the variable $z = f(x_0 + su_1, y_0 + su_2)$ is a function of the parameter s . The value of the derivative $\frac{dz}{ds}$ at $s = 0$ then gives an instantaneous rate of change of $f(x, y)$ with respect to distance from (x_0, y_0) in the direction of $\mathbf{u}$.

13.6.1 DEFINITION If $f(x, y)$ is a function of x and y , and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then the *directional derivative of f in the direction of $\mathbf{u}$* at (x_0, y_0) is denoted by $D_{\mathbf{u}}f(x_0, y_0)$ and is defined by

$$D_{\mathbf{u}}f(x_0, y_0) = \frac{d}{ds} [f(x_0 + su_1, y_0 + su_2)]_{s=0} \quad (2)$$

provided this derivative exists.

13.6.2 DEFINITION If $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a unit vector, and if $f(x, y, z)$ is a function of x , y , and z , then the *directional derivative of f in the direction of $\mathbf{u}$* at (x_0, y_0, z_0) is denoted by $D_{\mathbf{u}}f(x_0, y_0, z_0)$ and is defined by

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = \frac{d}{ds} [f(x_0 + su_1, y_0 + su_2, z_0 + su_3)]_{s=0} \quad (3)$$

provided this derivative exists.

13.6.3 THEOREM

(a) If $f(x, y)$ is differentiable at (x_0, y_0) , and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j}$ is a unit vector, then the directional derivative $D_{\mathbf{u}}f(x_0, y_0)$ exists and is given by

$$D_{\mathbf{u}}f(x_0, y_0) = f_x(x_0, y_0)u_1 + f_y(x_0, y_0)u_2 \quad (4)$$

(b) If $f(x, y, z)$ is differentiable at (x_0, y_0, z_0) , and if $\mathbf{u} = u_1\mathbf{i} + u_2\mathbf{j} + u_3\mathbf{k}$ is a unit vector, then the directional derivative $D_{\mathbf{u}}f(x_0, y_0, z_0)$ exists and is given by

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = f_x(x_0, y_0, z_0)u_1 + f_y(x_0, y_0, z_0)u_2 + f_z(x_0, y_0, z_0)u_3 \quad (5)$$

13.6.4 DEFINITION

(a) If f is a function of x and y , then the *gradient of f* is defined by

$$\nabla f(x, y) = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}$$

(b) If f is a function of x , y , and z , then the *gradient of f* is defined by

$$\nabla f(x, y, z) = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k}$$

Definition: The Dot product of two vectors $a = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ and $b = b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}$ is given by

$$a \cdot b = (a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}) \cdot (b_1\mathbf{i} + b_2\mathbf{j} + b_3\mathbf{k}) = a_1b_1 + a_2b_2 + a_3b_3$$

The symbol $\nabla = \frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k}$ (read “del”) is an inverted delta. (It is sometimes called a “nabla” because of its similarity in form to an ancient Hebrew ten-stringed harp of that name.) Formulas for directional derivatives can now be written as

$$D_{\mathbf{u}}f(x_0, y_0) = \nabla f(x_0, y_0) \cdot \mathbf{u}$$

$$D_{\mathbf{u}}f(x_0, y_0, z_0) = \nabla f(x_0, y_0, z_0) \cdot \mathbf{u}$$

Note that the gradient of f is

$$\begin{aligned} \nabla f(x, y, z) &= \left(\frac{\partial}{\partial x} \mathbf{i} + \frac{\partial}{\partial y} \mathbf{j} + \frac{\partial}{\partial z} \mathbf{k} \right) f(x, y, z) \\ &= \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j} + \frac{\partial f}{\partial z} \mathbf{k} = f_x(x, y, z)\mathbf{i} + f_y(x, y, z)\mathbf{j} + f_z(x, y, z)\mathbf{k} \end{aligned}$$

Definitions:

(a) A unit vector is a vector of length 1.

(b) The length of the vector $a = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ is denoted by $\|a\|$ and is given by

$$\|a\| = \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

(c) The unit vector in the direction of the vector $a = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ is given by

$$u = \frac{a}{\|a\|} = \frac{a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}}{\sqrt{a_1^2 + a_2^2 + a_3^2}}$$

Note that $\|u\| = \left\| \frac{a}{\|a\|} \right\| = \frac{1}{\|a\|} \|a\| = 1$.

PROPERTIES OF THE GRADIENT

The gradient is not merely a notational device to simplify the formula for the directional derivative; we will see that the length and direction of the gradient ∇f provide important information about the function f and the surface $z = f(x, y)$. For example,

- 1) At (x, y) , the surface $z = f(x, y)$ has its maximum slope in the direction of the gradient, and the maximum slope is $\|\nabla f(x, y)\|$. That is, the function $f(x, y)$ increases most rapidly in the direction of its gradient.
- 2) At (x, y) , the surface $z = f(x, y)$ has its minimum slope in the direction that is opposite to the gradient, and the minimum slope is $-\|\nabla f(x, y)\|$. That is, the function $f(x, y)$ decreases most rapidly in the direction opposite to its gradient.

13.6.5 THEOREM *Let f be a function of either two variables or three variables, and let P denote the point $P(x_0, y_0)$ or $P(x_0, y_0, z_0)$, respectively. Assume that f is differentiable at P .*

- (a) *If $\nabla f = \mathbf{0}$ at P , then all directional derivatives of f at P are zero.*
- (b) *If $\nabla f \neq \mathbf{0}$ at P , then among all possible directional derivatives of f at P , the derivative in the direction of ∇f at P has the largest value. The value of this largest directional derivative is $\|\nabla f\|$ at P .*
- (c) *If $\nabla f \neq \mathbf{0}$ at P , then among all possible directional derivatives of f at P , the derivative in the direction opposite to that of ∇f at P has the smallest value. The value of this smallest directional derivative is $-\|\nabla f\|$ at P .*

13.6.6 THEOREM *Assume that $f(x, y)$ has continuous first-order partial derivatives in an open disk centered at (x_0, y_0) and that $\nabla f(x_0, y_0) \neq \mathbf{0}$. Then $\nabla f(x_0, y_0)$ is normal to the level curve of f through (x_0, y_0) .*

Type of Questions from this section:

1. Finding directional derivatives in the direction of an arbitrary vector a .
2. Finding directional derivatives in the direction of a unit vector u .
3. Finding gradient vectors of functions.
4. Find a unit vector in the direction in which f increases/decreases most rapidly at P , and find the rate of change of f at P in that direction.
5. Find the directional derivative of f at P in the direction of a vector making the angle θ with the positive x -axis in counterclockwise direction. [2D Domain].

Definition: If a vector v makes an angle θ with the positive x -axis in the counterclockwise direction, then a unit vector in the direction of the vector v is given by $\mathbf{u} = \cos \theta \mathbf{i} + \sin \theta \mathbf{j}$.

EXERCISES

Example 1 Let $f(x, y) = 3x^2y + 7$. Find and interpret $D_u f(1, 2)$ for the unit vector $u = \frac{\sqrt{3}}{2}i + \frac{1}{2}j$.

Solution: Given $f(x, y) = 3x^2y + 7$.

We know that the directional derivation of $f(x, y)$ at the point $(1, 2)$ in the direction $u = u_1i + u_2j$ is given by

$$D_u f(1, 2) = f_x(1, 2)u_1 + f_y(1, 2)u_2 \dots (1)$$

Here, $f_x(x, y) = 6xy$, $f_y(x, y) = 3x^2$.

At $(1, 2)$: $f_x(1, 2) = 12$, $f_y(1, 2) = 3$

Given $u = \frac{\sqrt{3}}{2}i + \frac{1}{2}j = u_1i + u_2j \Rightarrow u_1 = \frac{\sqrt{3}}{2}$, $u_2 = \frac{1}{2}$.

Then from (1): the directional derivation of $f(x, y)$ at the point $(1, 2)$ in the direction u is given by

$$D_u f(1, 2) = f_x(1, 2)u_1 + f_y(1, 2)u_2$$

$$= 12\left(\frac{\sqrt{3}}{2}\right) + 3\left(\frac{1}{2}\right)$$

$$= 6\sqrt{3} + \frac{3}{2}$$

$$= \frac{3}{2}(4\sqrt{3} + 1)$$

Example 2 Let $f(x, y, z) = xyz^3 - xy$. Find and interpret $D_u f(1, 2, -1)$ for the vector

$$a = \frac{3}{2}i + \frac{1}{2}j - k.$$

Solution: Given $f(x, y, z) = xyz^3 - xy$ and vector $a = \frac{3}{2}i + \frac{1}{2}j - k$.

$$\text{Here, } ||a|| = \sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + (-1)^2} = \sqrt{\frac{9}{4} + \frac{1}{4} + 1} = \frac{\sqrt{14}}{2} \neq 1.$$

The unit vector in the direction of the given vector a : $u = \frac{\left(\frac{3}{2}i + \frac{1}{2}j - k\right)}{\frac{\sqrt{14}}{2}}$

$$\text{That is, } u = u_1i + u_2j + u_3k = \frac{3}{\sqrt{14}}i + \frac{1}{\sqrt{14}}j + \left(-\frac{2}{\sqrt{14}}\right)k$$

For $f(x, y, z) = xyz^3 - xy$; $f_x = yz^3 - y$, $f_y = xz^3 - x$, $f_z = 3xyz^2$

At $(1, 2, -1)$: $f_x = -4$, $f_y = -2$, $f_z = 6$

The directional derivative at $(1, 2, -1)$ in the direction of u :

$$\begin{aligned} D_u f(1, 2, -1) &= f_x(1, 2, -1)u_1 + f_y(1, 2, -1)u_2 + f_z(1, 2, -1)u_3 \\ &= (-4)\left(\frac{3}{\sqrt{14}}\right) + (-2)\left(\frac{1}{\sqrt{14}}\right) + 6\left(\frac{-2}{\sqrt{14}}\right) \\ &= \frac{-12}{\sqrt{14}} - \frac{2}{\sqrt{14}} - \frac{12}{\sqrt{14}} \\ &= -\frac{26}{\sqrt{14}} \end{aligned}$$

Example 3 Find the directional derivative of $f(x, y) = e^{xy}$ at $(-2, 0)$ in the direction of the unit vector that makes an angle of $\frac{\pi}{3}$ with the positive x -axis.

Solution: Given $f(x, y) = e^{xy}$; $P = P(-2, 0)$

➔ In the direction of the unit vector that makes an angle of $\theta = \frac{\pi}{3}$ with the positive x -axis.

The unit vector is $u = \cos\frac{\pi}{3}i + \sin\frac{\pi}{3}j = \frac{1}{2}i + \frac{\sqrt{3}}{2}j = u_1i + u_2j$

$$D_u f(-2, 0) = f_x(-2, 0)u_1 + f_y(-2, 0)u_2 \dots \dots \dots (1)$$

$$f_x(-2, 0) = 0, \quad f_y(-2, 0) = -2$$

$$D_u f(-2, 0) = \frac{1}{2}(0) + \frac{\sqrt{3}}{2}(-2) = -\sqrt{3}.$$

Note: Equation of a circle in the polar coordinate is: $r = k$, $k \geq 0$. Here $k = 0$ only at $(0, 0)$

Equation of the unit circle in the polar coordinate is: $r = 1$.

Points are given by $(x, y) = (\cos \theta, \sin \theta)$.

Definition: If a vector v makes an angle θ with the positive x -axis in the anti-clockwise direction, then a unit vector in the direction of the vector v is given by $u = \cos \theta i + \sin \theta j$.

Example 4 Find the directional derivative of $f(x, y, z) = x^2y - yz^3 + z$ at the point $(1, -2, 0)$ in the direction of the vector $a = 2i + j - 2k$. Homework

Example 5 Let $f(x, y) = x^2 e^y$. Find the maximum value of a directional derivative at $(-2, 0)$, and find the unit vector in the direction in which the maximum value occurs.

Solution: We know that, at (x, y) , the surface $z = f(x, y)$ has its maximum slope in the direction of the gradient, and the maximum slope is $\|\nabla f(x, y)\|$. That is, the function $f(x, y)$ increases most rapidly in the direction of its gradient.

The maximum slope in the direction of the gradient vector

(a) The maximum value of a directional derivative is $\|\nabla f(-2, 0)\| = ?$

(b) The unit vector in that direction: $u = \frac{\nabla f(-2, 0)}{\|\nabla f(-2, 0)\|}$.

Complete: Write the complete solution.

Answer: (Example 5)

Maximum value $= \sqrt{32}$, unit vector in that direction $u = \frac{1}{\sqrt{32}}(-4i + 4j) = \frac{1}{\sqrt{2}}(-i + j)$

Example 6

(a) Find the directional derivative of $f(x, y) = e^{xy}$ at $(-2, 0)$ in the direction of the negative y -axis. Section-11.

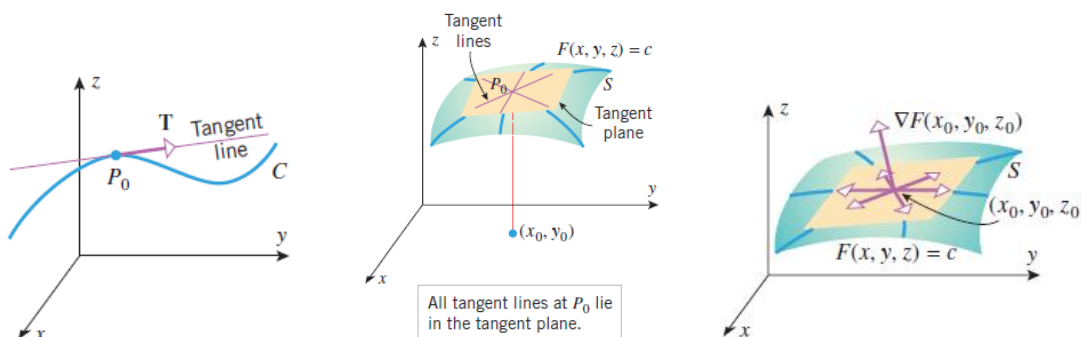
Hint: What is the unit vector that is in the direction of the negative y -axis? What is the angle θ that is between the positive x -axis and the negative y -axis in anti-clockwise direction?

SECTION 13.7 TANGENT PLANES AND NORMAL VECTORS

To Do:

- (1) Finding the equation of a tangent plane at some point
- (2) Finding the normal vector
- (3) Finding the parametric equations of normal lines
- (4) Finding angles between two planes/surfaces.

TANGENT PLANES AND NORMAL VECTORS TO LEVEL SURFACES $F(x, y, z) = c$



13.7.1 DEFINITION Assume that $F(x, y, z)$ has continuous first-order partial derivatives and that $P_0(x_0, y_0, z_0)$ is a point on the level surface $S: F(x, y, z) = c$. If $\nabla F(x_0, y_0, z_0) \neq \mathbf{0}$, then $\mathbf{n} = \nabla F(x_0, y_0, z_0)$ is a **normal vector** to S at P_0 and the **tangent plane** to S at P_0 is the plane with equation

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

Recall: $k = w = f(x, y, z) \leftarrow$ Level surface.

Level Surface $\rightarrow$ equations of three variables.

For $z = f(x, y) \rightarrow z - f(x, y) = 0 \therefore F(x, y, z) = z - f(x, y) = 0$.

Note that the gradient of a function $F(x, y, z)$

$$\mathbf{n} = \nabla F(x, y, z) = \langle F_x(x, y, z), F_y(x, y, z), F_z(x, y, z) \rangle = F_x(x, y, z)\mathbf{i} + F_y(x, y, z)\mathbf{j} + F_z(x, y, z)\mathbf{k}.$$

The line through the point P_0 parallel to the normal vector $\mathbf{n}$ is perpendicular to the tangent plane. We will call this the **normal line**, or sometimes more simply the **normal** to the surface $F(x, y, z) = c$ at P_0 .

The **parametric equations of normal lines** at the point $P_0 = (x_0, y_0, z_0)$ is given by

$$x = x_0 + F_x(x_0, y_0, z_0)t, \quad y = y_0 + F_y(x_0, y_0, z_0)t, \quad z = z_0 + F_z(x_0, y_0, z_0)t$$

TANGENT PLANES TO SURFACES OF THE FORM $z = f(x, y)$, Rewrite: $z - f(x, y) = 0$

To find a tangent plane to a surface of the form $z = f(x, y)$, we can use equation of the tangent plane with the function $F(x, y, z) = z - f(x, y)$. The level surface is given by

$$F(x, y, z) = z - f(x, y) = 0.$$

Definition:

If θ is the angle between two vectors $a = a_1i + a_2j + a_3k$ and $b = b_1i + b_2j + b_3k$, then θ is given by

$$\cos \theta = \frac{a \cdot b}{\|a\| \|b\|}, \text{ that is, } \theta = \cos^{-1} \left(\frac{a \cdot b}{\|a\| \|b\|} \right).$$

Note: The length of the vector $a = a_1i + a_2j + a_3k$ is $\|a\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$

Summary:

Let the point $P_0 = P_0(x_0, y_0, z_0)$ and the level surface $S: F(x, y, z) = C$.

(1) The equation of the tangent plane at (x_0, y_0, z_0) :

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0 \quad \dots (1)$$

(2) The Normal Vector at (x_0, y_0, z_0) :

$$\nabla F(x_0, y_0, z_0) = F_x(x_0, y_0, z_0)i + F_y(x_0, y_0, z_0)j + F_z(x_0, y_0, z_0)k \quad \dots (2)$$

(3) The parametric equations of normal lines at (x_0, y_0, z_0) :

$$x = F_x(x_0, y_0, z_0)t + x_0, \quad y = F_y(x_0, y_0, z_0)t + y_0, \quad z = F_z(x_0, y_0, z_0)t + z_0 \quad \dots (3)$$

(4) If θ is the angle between two vectors $a = a_1i + a_2j + a_3k$ and $b = b_1i + b_2j + b_3k$, then

θ is given by

$$\cos \theta = \frac{a \cdot b}{\|a\| \|b\|}, \text{ that is, } \theta = \cos^{-1} \left(\frac{a \cdot b}{\|a\| \|b\|} \right) \quad \dots (4)$$

Example: 1 Consider the Ellipsoid $4x^2 + y^2 + 3z^2 = 20$.

- (a) Find the equation of a tangent plane to the Ellipsoid at point $(1, 2, -2)$.
- (b) Find the normal vector at $(1, 2, -2)$.
- (c) Find the parametric equations of normal lines at $(1, 2, -2)$.
- (d) (i) Find the angle between the tangent plane to the surface at $(\sqrt{2}, 0, -2)$ and the xz -plane.
 (ii) Find the angle between the tangent plane to the surface at $(1, 2, -2)$ and the plane $2x - 3y = -z - 6$.
 (iii) Find the angle between surfaces $4x^2 + y^2 + 3z^2 = 20$ and $4x^2 + 3y^2 + z^2 = 20$ at $(1, 2, -2)$.

$$\left[\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \rightarrow \text{Ellipse} \right]$$



Solution:

Set $F(x, y, z) = 4x^2 + y^2 + 3z^2$. Then we get the level surface, $S: F(x, y, z) = 20$.

Now, $F(x, y, z) = 4x^2 + y^2 + 3z^2$

$$F_x(x, y, z) = 8x, \quad F_y(x, y, z) = 2y, \quad F_z(x, y, z) = 6z$$

$$\text{At } (1, 2, -2): \quad F_x(1, 2, -2) = 8, \quad F_y(1, 2, -2) = 4, \quad F_z(1, 2, -2) = -12.$$

(a) The equation of the tangent plane at $(x_0, y_0, z_0) = (1, 2, -2)$:

$$F_x(x_0, y_0, z_0)(x - x_0) + F_y(x_0, y_0, z_0)(y - y_0) + F_z(x_0, y_0, z_0)(z - z_0) = 0$$

$$\Rightarrow F_x(1, 2, -2)(x - 1) + F_y(1, 2, -2)(y - 2) + F_z(1, 2, -2)(z + 2) = 0$$

$$\Rightarrow 8(x - 1) + 4(y - 2) - 12(z + 2) = 0$$

$$\Rightarrow 8x + 4y - 12z = 40$$

$$\Rightarrow 4(2x + y - 3z) = 40$$

$$\therefore 2x + y - 3z = 10.$$

(b) The normal vector at $(1, 2, -2)$: $n = F_x(1, 2, -2)i + F_y(1, 2, -2)j + F_z(1, 2, -2)k$

Therefore, $n = 8i + 4j - 12k$.

(c) The parametric equations of normal lines at $(x_0, y_0, z_0) = (1, 2, -2)$:

$$x = F_x(x_0, y_0, z_0)t + x_0 \Rightarrow x = 8t + 1$$

$$y = F_y(x_0, y_0, z_0)t + y_0 \Rightarrow y = 4t + 2$$

$$z = F_z(x_0, y_0, z_0)t + z_0 \Rightarrow z = -12t - 2$$

(d) (i) Find the angle between the tangent plane to the surface at $(\sqrt{2}, 0, -2)$ and the xz -plane.

(ii) Find the angle between the tangent plane to the surface at $(1, 2, -2)$ and the plane

$$2x - 3y = -z - 6.$$

(iii) Find the angle between surfaces $4x^2 + y^2 + 3z^2 = 20$ and $4x^2 + 3y^2 + z^2 = 20$ at $(1, 2, -2)$.

(d) (i) Find the angle between the tangent plane to the surface at $(\sqrt{2}, 0, -2)$ and the xz -plane.

Solution: Given $4x^2 + y^2 + 3z^2 = 20$.

Set $F(x, y, z) = 4x^2 + y^2 + 3z^2$. Then we get the level surface, $S: F(x, y, z) = 20$.

$$\text{Now, } F(x, y, z) = 4x^2 + y^2 + 3z^2$$

$$F_x(x, y, z) = 8x, \quad F_y(x, y, z) = 2y, \quad F_z(x, y, z) = 6z$$

$$\text{At } (\sqrt{2}, 0, -2): F_x(\sqrt{2}, 0, -2) = 8\sqrt{2}, \quad F_y(\sqrt{2}, 0, -2) = 0, \quad F_z(\sqrt{2}, 0, -2) = -12.$$

Here the normal vector the tangent plane at $(1, 0, -2)$ is

$$n_1 = F_x(\sqrt{2}, 0, -2)i + F_y(\sqrt{2}, 0, -2)j + F_z(\sqrt{2}, 0, -2)k$$

$$\therefore n_1 = 8\sqrt{2}i + 0j - 12k.$$

Now, the equation of the xz –plane: $y = 0$. Set $G(x, y, z) = y$.

The level surface given by the xz –plane: $G(x, y, z) = 0$.

Taking partial derivatives of $G(x, y, z) = y$:

$$G_x(x, y, z) = 0, \quad G_y(x, y, z) = 1, \quad G_z(x, y, z) = 0$$

$$G_x = 0, \quad G_y = 1, \quad G_z = 0.$$

The normal vector to the xz –plane: $n_2 = 0i + j + 0k$.

Now, the angle θ between the vectors n_1 and n_2 is:

$$\cos \theta = \frac{n_1 \cdot n_2}{\|n_1\| \|n_2\|} = \frac{(8\sqrt{2}i + 0j - 12k) \cdot (j)}{\sqrt{(8\sqrt{2})^2 + 0^2 + (-12)^2} \sqrt{1^2}} = \frac{0}{\sqrt{64(2) + 144}} = 0$$
$$\Rightarrow \theta = \cos^{-1}(0) \approx \frac{\pi}{2} \text{ rad}$$

[Calculator must be in radian mode]

(d/ ii) Find the angle between the paraboloid and the plane $2x - 3y = -z - 6$ at $(1, 2, -2)$.

The normal vector to the paraboloid at the point $(1, 2, -2)$: $n_1 = 8i + 4j - 12k$.

Second surface: $G(x, y, z) = 2x - 3y + z = -6$.

Here, $n_2 = 2i - 3j + k$

(d/iii) Find the angle between surfaces $4x^2 + y^2 + 3z^2 = 20$ and $4x^2 + 3y^2 + z^2 = 20$ at $(1, 2, -2)$.

The normal vector to the paraboloid at the point $(1, 2, -2)$:

$$n_1 = 8i + 4j - 12k.$$

The normal vector to the surface at the point $(1, 2, -2)$ to the surface $4x^2 + 3y^2 + z^2 = 20$:

$$n_2 = 8i + 12j - 4k$$

$$\cos \theta = \frac{n_1 \cdot n_2}{\|n_1\| \|n_2\|} \rightarrow \theta \approx 0.78 \text{ rad}$$

Homework

Example: 2 Consider $f(x, y) = \sqrt{x^2 + y^2}$

- (a) Find the equation of a tangent plane at point $(3, 4, 5)$.
- (b) Find the normal vector at $(3, 4, 5)$.
- (c) Find the parametric equations of normal lines at $(3, 4, 5)$.
- (d) (i) Find the angle between the tangent plane at $(3, 4, 5)$ and the xy -plane.
 (ii) Find the angle between the tangent plane at $(3, 4, 5)$ and the plane $z = -x + 2y$.

Solution: Given $f(x, y) = \sqrt{x^2 + y^2}$.

Set $z = f(x, y)$. Then $z = \sqrt{x^2 + y^2}$, i.e., $z - \sqrt{x^2 + y^2} = 0$

Then $z - f(x, y) = 0 \Rightarrow z - \sqrt{x^2 + y^2} = 0$.

Set $F(x, y, z) = z - \sqrt{x^2 + y^2}$. The level surface is $S: F(x, y, z) = 0$.

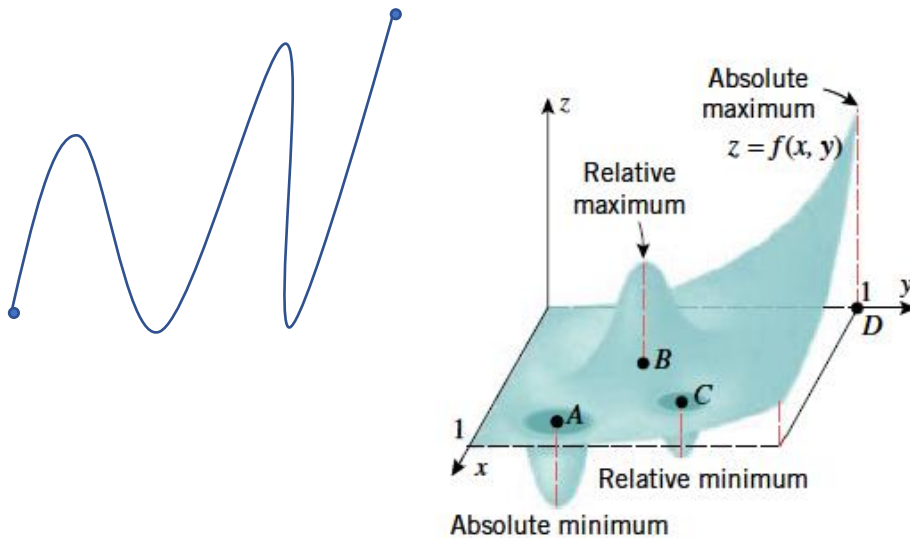
Now, $F(x, y, z) = z - \sqrt{x^2 + y^2}$. **Please complete!**

Exercise 3.

- (a) Find the angle between the surfaces $3x^2 + 4y^2 + 6z^2 = 20$ and $4x^2 + 3y^2 + z^2 = 20$ at the point $(1, -1, 0)$. **Section -11, section-12, section -7**

$$\cos \theta = \frac{n_1 \cdot n_2}{\|n_1\| \|n_2\|}$$

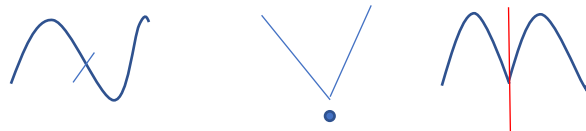
SECTION 13.8 MAXIMA AND MINIMA OF FUNCTIONS OF TWO VARIABLES



13.8.1 DEFINITION A function f of two variables is said to have a *relative maximum* at a point (x_0, y_0) if there is a disk centered at (x_0, y_0) such that $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) that lie inside the disk, and f is said to have an *absolute maximum* at (x_0, y_0) if $f(x_0, y_0) \geq f(x, y)$ for all points (x, y) in the domain of f .

13.8.2 DEFINITION A function f of two variables is said to have a *relative minimum* at a point (x_0, y_0) if there is a disk centered at (x_0, y_0) such that $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) that lie inside the disk, and f is said to have an *absolute minimum* at (x_0, y_0) if $f(x_0, y_0) \leq f(x, y)$ for all points (x, y) in the domain of f .

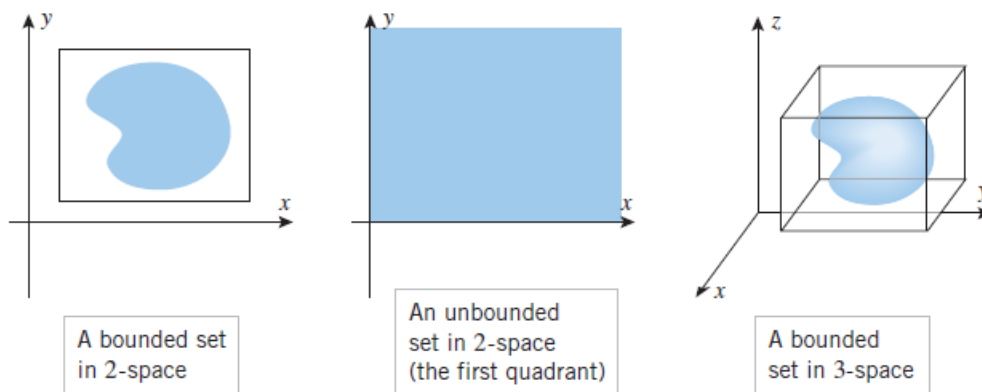
Recall $\rightarrow$ MAT-120:



For $y = f(x) : (c, f(c))$ critical point $\leftrightarrow$ Either $f'(c) = 0$ Or $f'(c)$ Undefined.

Theorem: If any Relative (Local) Max/Min occurs, then it must appear at some critical point.

BOUNDED SETS



THE EXTREME-VALUE THEOREM

For functions of one variable that are continuous on a closed interval, the Extreme-Value Theorem answered the existence question for absolute extrema. The following theorem, which we state without proof, is the corresponding result for functions of two variables.

13.8.3 THEOREM (Extreme-Value Theorem) *If $f(x, y)$ is continuous on a closed and bounded set R , then f has both an absolute maximum and an absolute minimum on R .*

For the function $y = f(x)$, if $y = f(x)$ is continuous on the closed interval $[a, b]$, then $y = f(x)$ has absolute maximum and minimum on that interval $[a, b]$.

Note: $[a, b]$ is bounded because $[a, b] \subset (a - \varepsilon, b + \varepsilon) \subset [a - \varepsilon, b + \varepsilon]$.

FINDING RELATIVE EXTREMA

Recall:

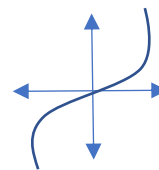
If a function g of one variable has a relative extremum at a point x_0 where g is differentiable, then $g'(x_0) = 0$ or $g'(x_0)$ is undefined.

To obtain the analog of this result for functions of two variables, suppose that $f(x, y)$ has a relative maximum at a point (x_0, y_0) and that the partial derivatives of f exist at (x_0, y_0) . It seems plausible geometrically that the traces of the surface $z = f(x, y)$ on the planes $x = x_0$ and $y = y_0$ have horizontal tangent lines at (x_0, y_0) , so

$$f_x(x_0, y_0) = 0 \quad \text{and} \quad f_y(x_0, y_0) = 0.$$

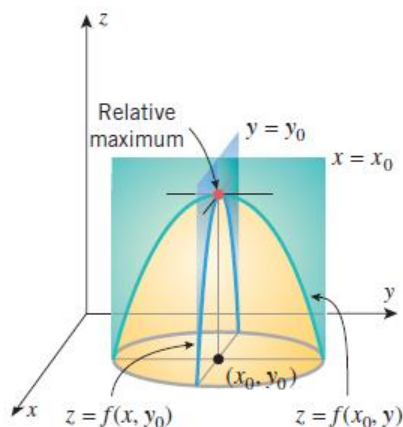
The same conclusion holds if f has a relative minimum at (x_0, y_0) .

$$y = x^3$$



13.8.4 THEOREM If f has a relative extremum at a point (x_0, y_0) , and if the first-order partial derivatives of f exist at this point, then

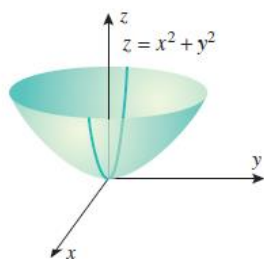
$$f_x(x_0, y_0) = 0 \quad \text{and} \quad f_y(x_0, y_0) = 0$$



Recall: The *critical points* of a function f of one variable are those values of x in the domain of f at which $f'(x) = 0$ or f is not differentiable.

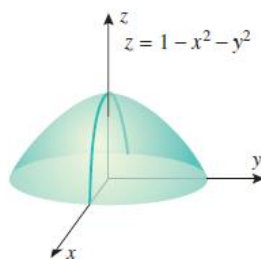
The following definition is the analog for functions of two variables.

13.8.5 DEFINITION A point (x_0, y_0) in the domain of a function $f(x, y)$ is called a *critical point* of the function if $f_x(x_0, y_0) = 0$ and $f_y(x_0, y_0) = 0$ or if one or both partial derivatives do not exist at (x_0, y_0) .



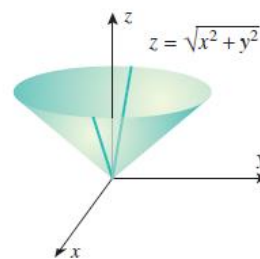
$f_x(0, 0) = f_y(0, 0) = 0$
relative and absolute min at $(0, 0)$

(a)



$f_x(0, 0) = f_y(0, 0) = 0$
relative and absolute max at $(0, 0)$

(b)



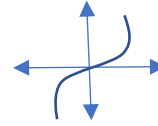
$f_x(0, 0)$ and $f_y(0, 0)$ do not exist
relative and absolute min at $(0, 0)$

(c)

Given, $z = f(x, y) = \sqrt{x^2 + y^2}$. Then $f_x = \frac{\partial z}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}}$ and $f_y = \frac{\partial z}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}}$

[Either (1) $f_x = 0$ and $f_y = 0$

Or (2) f_x or f_y or both are undefined.]



THE SECOND PARTIAL DERIVATIVES TEST

For functions of one variable the second derivative test was used to determine the behavior of a function at a critical point. The following theorem, which is usually proved in advanced calculus, is the analog of that theorem for functions of two variables.

13.8.6 THEOREM (The Second Partial Test) Let f be a function of two variables with continuous second-order partial derivatives in some disk centered at a critical point (x_0, y_0) , and let

$$D = f_{xx}(x_0, y_0)f_{yy}(x_0, y_0) - f_{xy}^2(x_0, y_0)$$

(a) If $D > 0$ and $f_{xx}(x_0, y_0) > 0$, then f has a relative minimum at (x_0, y_0) .

(b) If $D > 0$ and $f_{xx}(x_0, y_0) < 0$, then f has a relative maximum at (x_0, y_0) .

(c) If $D < 0$, then f has a saddle point at (x_0, y_0) .

(d) If $D = 0$, then no conclusion can be drawn.

Given $z = f(x, y)$. Set $D = f_{xx}(x_0, y_0) \cdot f_{yy}(x_0, y_0) - [f_{xy}(x_0, y_0)]^2$. Examine the sign of D and $f_{xx}(x_0, y_0)$.

Example 1/A Locate all relative extrema and saddle points of $f(x, y) = 3x^2 - 2xy + y^2 - 8y$.

Solution: Given $f(x, y) = 3x^2 - 2xy + y^2 - 8y$.

Then $\frac{\partial f}{\partial x} = f_x = 6x - 2y$, $\frac{\partial f}{\partial y} = f_y = -2x + 2y - 8$

For the critical point, set

$$\Rightarrow 6x - 2y = 0 \dots \dots (1)$$

$$2x - 2y = -8 \dots \dots (2)$$

$$(1) - (2): 4x = 8, \quad i.e., x = 2$$

Plug in $x = 2$ in the equation (1): $12 - 2y = 0$, $i.e., y = 6$

Hence, the critical point is $(2, 6)$.

We have $f_x = 6x - 2y = 0$, $f_y = -2x + 2y - 8 = 0$

Now, $f_{xx} = 6$, $f_{yy} = 2$ and $f_{xy} = -2$ at each point in the domain.

$$\text{Set } D = f_{xx}(2, 6) \cdot f_{yy}(2, 6) - [f_{xy}(2, 6)]^2 = (6)(2) - (-2)^2 = 12 - 4 = 8 > 0$$

Since $D = 8 > 0$ and $f_{xx}(2,6) = 6 > 0$, hence the function $z = f(x, y)$ has a relative minimum at the critical point $(2, 6)$. And the minimum value is $f(2, 6) = 3(2)^2 - 2(2)(6) + (6)^2 - 8(6) = -24$.

The function does not have any relative maximum or any saddle point.

Example 1/B Locate all relative extrema and saddle points of $f(x, y) = x^3 + y^3 - 12x - 3y + 1$.

Solution: Given $f(x, y) = x^3 + y^3 - 12x - 3y + 1$.

$$f_x = 3x^2 - 12 \quad \& \quad f_y = 3y^2 - 3 \quad \dots (1)$$

For critical points, $f_x = 3x^2 - 12 = 0 \quad \& \quad f_y = 3y^2 - 3 = 0$

Now, $f_x = 3x^2 - 12 = 0 \rightarrow x^2 = 4 \quad \therefore x = \pm 2$

$$f_y = 3y^2 - 3 = 0 \rightarrow \therefore y = \pm 1$$

Critical Points: $(2, 1), (2, -1), (-2, 1), (-2, -1)$.

Also, $f_{xx} = 6x, \quad f_{yy} = 6y \quad \& \quad f_{xy} = 0$

At the point $(2, 1)$: $f_{xx} = 12, \quad f_{yy} = 6, \quad f_{xy} = 0$.

So, $D = f_{xx}f_{yy} - (f_{xy})^2 = (12)(6) - 0^2 = 72 > 0$. Since $f_{xx} = 12 > 0$, the function f has a relative minimum at the point $(2, 1)$, and the minimum value is $f(2, 1) = 2^3 + 1^3 - 12(2) - 3(1) + 1 = -17$.

At the point $(2, -1)$: $f_{xx} = 12, \quad f_{yy} = -6, \quad f_{xy} = 0 \rightarrow D = (12)(-6) - 0 = -72 < 0$.

Hence $(2, -1)$ is a saddle point.

Similarly, $(-2, 1)$ is a saddle point.

AND

the function f has a relative maximum at the point $(-2, -1)$, and the maximum value is $f(-2, -1) = -8 - 1 + 24 + 3 + 1 = 19$.

Example 2 Locate all relative extrema and saddle points of $f(x, y) = x^3 + y^2 - 8x + 2y$.

Solution: Given $f(x, y) = x^3 + y^2 - 8x + 2y$

Then $\frac{\partial f}{\partial x} = f_x = 3x^2 - 8, \quad \frac{\partial f}{\partial y} = f_y = 2y + 2$

For the critical point, set $\frac{\partial f}{\partial x} = f_x = 3x^2 - 8 = 0 \rightarrow (1) \quad , \quad \frac{\partial f}{\partial y} = f_y = 2y + 2 = 0 \rightarrow (2)$

From (2): $y = -1$.

From (1): $3x^2 = 8 \rightarrow x^2 = \frac{8}{3} \therefore x = \pm \sqrt{\frac{8}{3}}$.

Critical Points are: $\left(\sqrt{\frac{8}{3}}, -1\right)$, $\left(-\sqrt{\frac{8}{3}}, -1\right)$.

Now, $f_{xx} = 6x$, $f_{yy} = 2$ & $f_{xy} = 0$.

Set $D = f_{xx}(x_0, y_0) \cdot f_{yy}(x_0, y_0) - [f_{xy}(x_0, y_0)]^2$.

At the critical point $\left(\sqrt{\frac{8}{3}}, -1\right)$: $f_{xx}\left(\sqrt{\frac{8}{3}}, -1\right) = 6\sqrt{\frac{8}{3}} > 0$ & $D = 6\sqrt{\frac{8}{3}}(2) - 0^2 = 12\sqrt{\frac{8}{3}} > 0$.

Since $D > 0$ and $f_{xx} > 0$ at the critical point $\left(\sqrt{\frac{8}{3}}, -1\right)$, so the function f has a local minimum at $\left(\sqrt{\frac{8}{3}}, -1\right)$.

The minimum value is

$$\begin{aligned} f\left(\sqrt{\frac{8}{3}}, -1\right) &= \left(\sqrt{\frac{8}{3}}\right)^3 + (-1)^2 - 8\sqrt{\frac{8}{3}} + 2(-1) \\ &= \frac{8}{3}\sqrt{\frac{8}{3}} + 1 - 8\sqrt{\frac{8}{3}} - 2 \\ &= -\frac{16}{3}\sqrt{\frac{8}{3}} - 1 \end{aligned}$$

At the critical point $\left(-\sqrt{\frac{8}{3}}, -1\right)$:

$$f_{xx}\left(-\sqrt{\frac{8}{3}}, -1\right) = 6\left(-\sqrt{\frac{8}{3}}\right) = -6\sqrt{\frac{8}{3}} < 0 \text{ \& } D = \left(-6\sqrt{\frac{8}{3}}\right)(2) - 0^2 = -12\sqrt{\frac{8}{3}} < 0.$$

Since $D < 0$, so the point $\left(-\sqrt{\frac{8}{3}}, -1\right)$ is a saddle point.

There is NO relative maximum.

From equation (2), we get $y = -1$

From the equation (1): $x^2 = \frac{8}{3}$, i.e., $x = \pm \sqrt{\frac{8}{3}}$

Hence, the critical points are $\left(\sqrt{\frac{8}{3}}, -1\right)$, $\left(-\sqrt{\frac{8}{3}}, -1\right)$.

Now, $f_{xx} = 6x$, $f_{yy} = 2$ and $f_{xy} = 0$.

At the critical point $\left(\sqrt{\frac{8}{3}}, -1\right)$:

$$\text{Set } D = f_{xx}\left(\sqrt{\frac{8}{3}}, -1\right) \cdot f_{yy}\left(\sqrt{\frac{8}{3}}, -1\right) - \left[f_{xy}\left(\sqrt{\frac{8}{3}}, -1\right)\right]^2 = 6\sqrt{\frac{8}{3}}(2) - 0^2 = 12\sqrt{\frac{8}{3}} > 0$$

$$\text{Also, at } \left(\sqrt{\frac{8}{3}}, -1\right), f_{xx} = 6\sqrt{\frac{8}{3}} > 0.$$

Since $D > 0$ and $f_{xx}\left(\sqrt{\frac{8}{3}}, -1\right) > 0$, hence the function $z = f(x, y)$ has a relative minimum at the critical point $\left(\sqrt{\frac{8}{3}}, -1\right)$. And the minimum value is $f\left(\sqrt{\frac{8}{3}}, -1\right) = \frac{32\sqrt{6}}{9} - 1$.

At the critical point $\left(-\sqrt{\frac{8}{3}}, -1\right)$:

$$\text{Set } D = f_{xx}\left(-\sqrt{\frac{8}{3}}, -1\right) \cdot f_{yy}\left(-\sqrt{\frac{8}{3}}, -1\right) - \left[f_{xy}\left(-\sqrt{\frac{8}{3}}, -1\right)\right]^2 = 6\left(-\sqrt{\frac{8}{3}}\right)(2) - 0^2 = -12\sqrt{\frac{8}{3}} < 0$$

Since $D < 0$, hence the function $z = f(x, y)$ has a saddle point at $\left(-\sqrt{\frac{8}{3}}, -1\right)$.

The function does not have any relative maximum.

Example 3 Locate all relative extrema and saddle points of $f(x, y) = 3x^4 - 12xy + 3y^4$

Solution: Homework !!!

FINDING ABSOLUTE EXTREMA ON CLOSED AND BOUNDED SETS

If $f(x, y)$ is continuous on a closed and bounded set R , then the Extreme-Value Theorem guarantees the existence of an absolute maximum and an absolute minimum of f on R . These absolute extrema can occur either on the boundary of R or in the interior of R , but if an absolute extremum occurs in the interior, then it occurs at a critical point. Thus, we are led to the following procedure for finding absolute extrema:

13.8.7 THEOREM *If a function f of two variables has an absolute extremum (either an absolute maximum or an absolute minimum) at an interior point of its domain, then this extremum occurs at a critical point.*

How to Find the Absolute Extrema of a Continuous Function f of Two Variables on a Closed and Bounded Set R

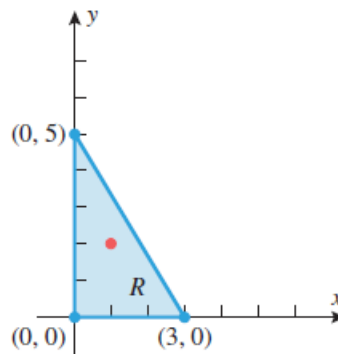
Step 1. Find the critical points of f that lie in the interior of R .

Step 2. Find all boundary points at which the absolute extrema can occur.

Step 3. Evaluate $f(x, y)$ at the points obtained in the preceding steps. The largest of these values is the absolute maximum and the smallest the absolute minimum.

Example 4 Find the absolute maximum and minimum values of $f(x, y) = 3xy - 6x - 3y + 7$ on the closed triangular region R with vertices $(0, 0)$, $(3, 0)$, and $(0, 5)$.

Solution:



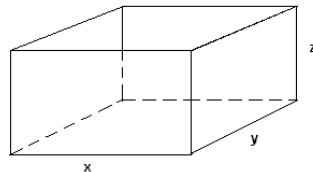
The line segment between $(0, 0)$ and $(3, 0)$:

The line segment between $(0, 0)$ and $(0, 5)$:

The line segment between $(0, 3)$ and $(0, 5)$:

Example 5 Determine the dimensions of a rectangular box, open at the top, having a volume of 32 ft^3 , and requiring the least amount of material for its construction.

Solution.



Let

x = length of the box (in feet)

y = width of the box (in feet)

z = height of the box (in feet)

A = surface area of the box (in square feet)

Surface area = $xy + 2yz + 2zx$ &

$$xyz = 32 \rightarrow z = \frac{32}{xy} \dots (1)$$

So, the surface area $A(x, y) = xy + 2y\left(\frac{32}{xy}\right) + 2\left(\frac{32}{xy}\right)x$

$$\therefore A(x, y) = xy + 64x^{-1} + 64y^{-1}$$

Task !!! $\rightarrow$ Finding the local minimum using second partial test.

.....

Let

x = length of the box (in feet)

y = width of the box (in feet)

z = height of the box (in feet)

S = surface area of the box (in square feet)

$$\text{Given, } xyz = 32 \Rightarrow z = \frac{32}{xy} \dots \dots \dots (1)$$

$$\text{Now } S = xy + 2yz + 2zx = xy + 2y \frac{32}{xy} + 2x \frac{32}{xy} = xy + 64 \frac{1}{x} + 64 \frac{1}{y}$$

The surface area of the box (in square feet) is given by the function:

$$S(x, y) = xy + 64 \frac{1}{x} + 64 \frac{1}{y}$$

GOAL: We need to find the minimum value of $S(x, y)$.

$$\text{Here, } s_x = y - \frac{64}{x^2}, \quad s_y = x - \frac{64}{y^2}, \quad S_{xx} = \frac{128}{x^3}, \quad S_{yy} = \frac{128}{y^3}, \quad S_{xy} = 1$$

$$\text{For the critical points: } s_x = y - \frac{64}{x^2} = 0, \quad s_y = x - \frac{64}{y^2} = 0$$

We get $y = \frac{64}{x^2} \dots \dots \dots (2)$ and then

$$x - \frac{64}{\left(\frac{64}{x^2}\right)^2} = 0 \Rightarrow x - \frac{64x^4}{64^2} = 0 \Rightarrow x - \frac{x^4}{64} = 0$$

$$\Rightarrow 64x - x^4 = 0 \Rightarrow x(64 - x^3) = 0, \text{ that is, } x = 0, 4. \text{ Hence, } x = 4.$$

From (2): $y = 4$.

Hence, we get only one critical point (4,4).

At (4,4):

$$S_{xx}(4,4) = \frac{128}{4^3} = 2, \quad S_{yy}(4,4) = \frac{128}{4^3} = 2, \quad S_{xy}(4,4) = 1$$

Now,

$$D = S_{xx}(4,4) S_{yy}(4,4) - \left(S_{xy}(4,4)\right)^2 = 2(2) - 1 = 3 > 0 \text{ and } S_{xx}(4,4) = 2 > 0.$$

Hence the surface area of the given open top box is minimum at (4,4). Here $z = \frac{32}{xy} = \frac{32}{16} = 2$.

Therefore, the dimension of the box:

Length of the box is 4 feet

Width of the box is 4 feet

Height of the box is 2 feet